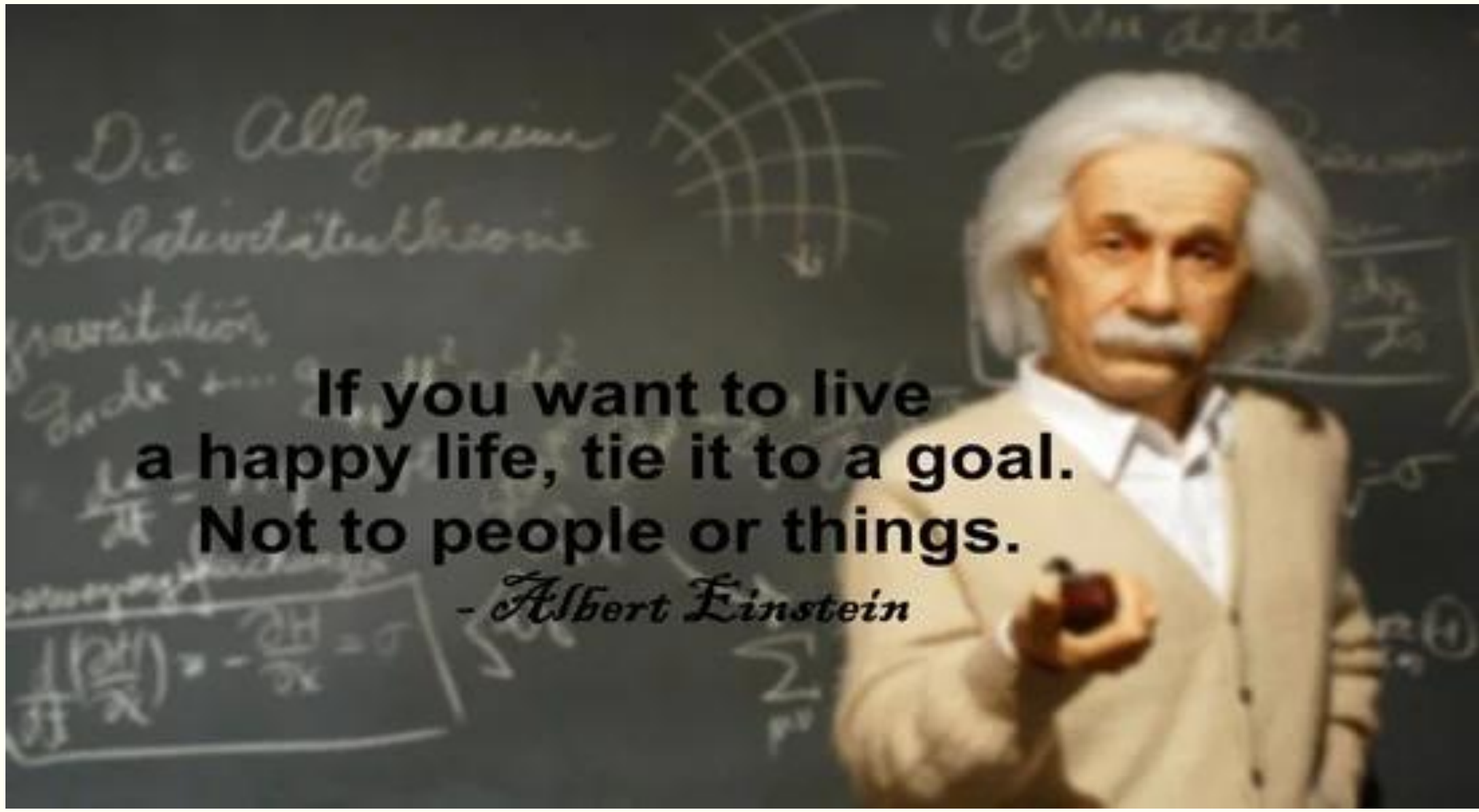


A photograph of Albert Einstein standing in front of a chalkboard filled with his work. He is holding an apple in his right hand. The chalkboard contains German text and mathematical equations related to his theory of relativity.

**If you want to live
a happy life, tie it to a goal.
Not to people or things.**

- Albert Einstein

MODULE NO:-5

Time series Forecasting

EPCGroup Microsoft

Time-Series Forecasting



Module No: -5 Time Series Forecasting

5.1 Taxonomy of Time Series Forecasting methods,

Time Series Decomposition

5.2 Smoothing Methods:

Average method,

Moving Average smoothing,

Time series analysis using linear regression,

ARIMA Model,

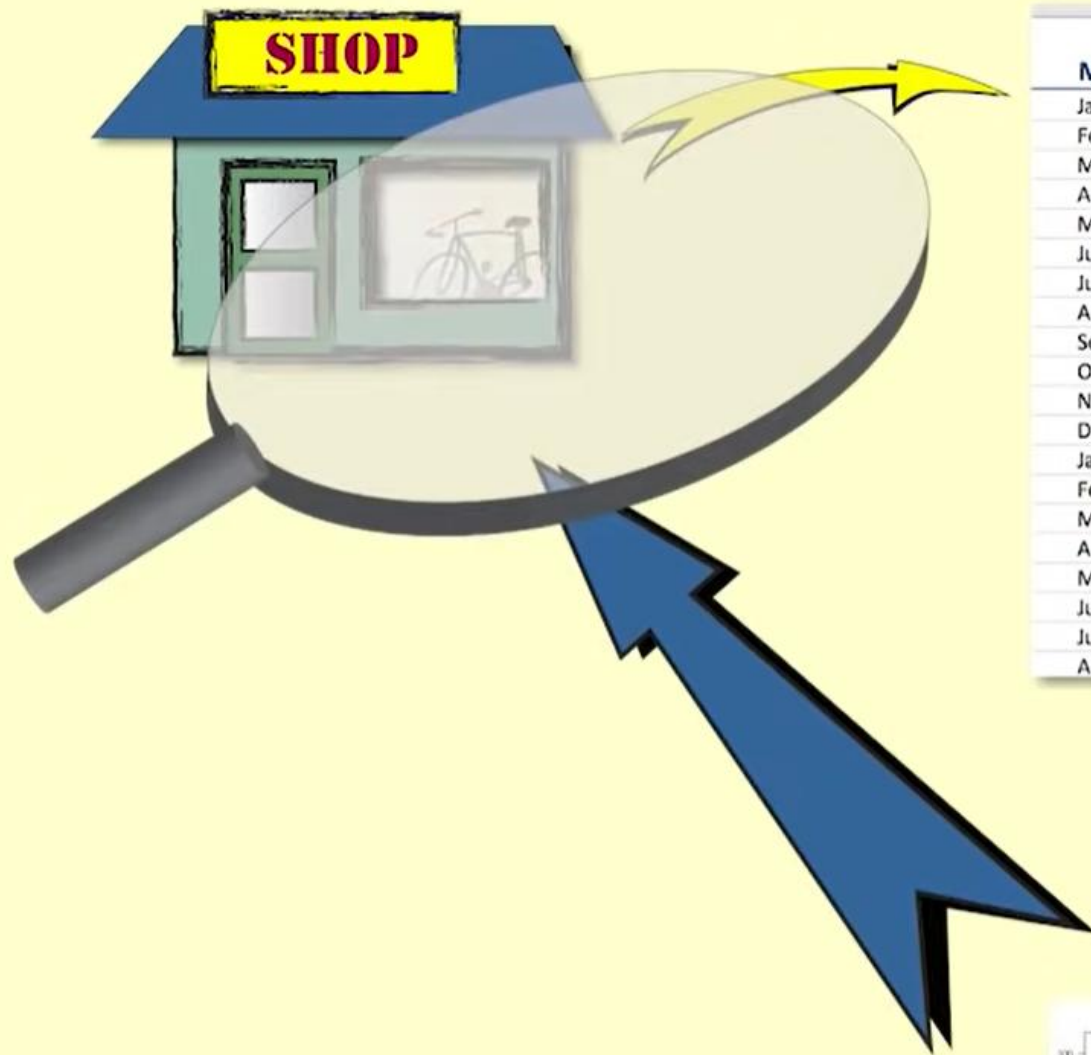
5.3 Performance Evaluation:

Mean Absolute Error,

Root Mean Square Error,

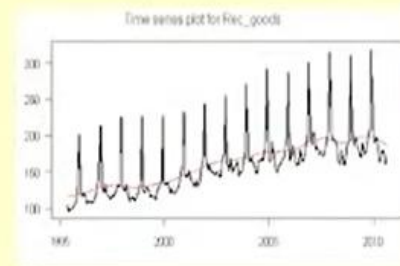
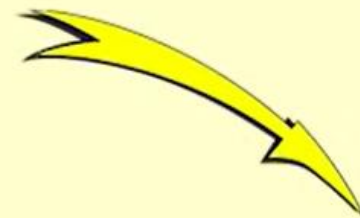
Mean Absolute Percentage Error,

Mean Absolute Scaled Error



Month	Recreational goods
Jan 2000	134.1
Feb 2000	133
Mar 2000	136.2
Apr 2000	120.4
May 2000	126.2
Jun 2000	119.8
Jul 2000	122.5
Aug 2000	122.3
Sep 2000	131.2
Oct 2000	138.7
Nov 2000	158.4
Dec 2000	232.2
Jan 2001	143.1
Feb 2001	142.4
Mar 2001	152.6
Apr 2001	139.9
May 2001	135.6
Jun 2001	129.4
Jul 2001	138.5
Aug 2001	130.2

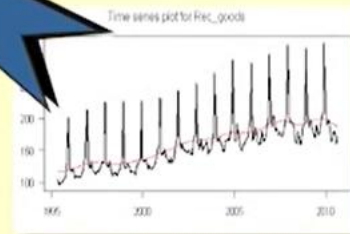
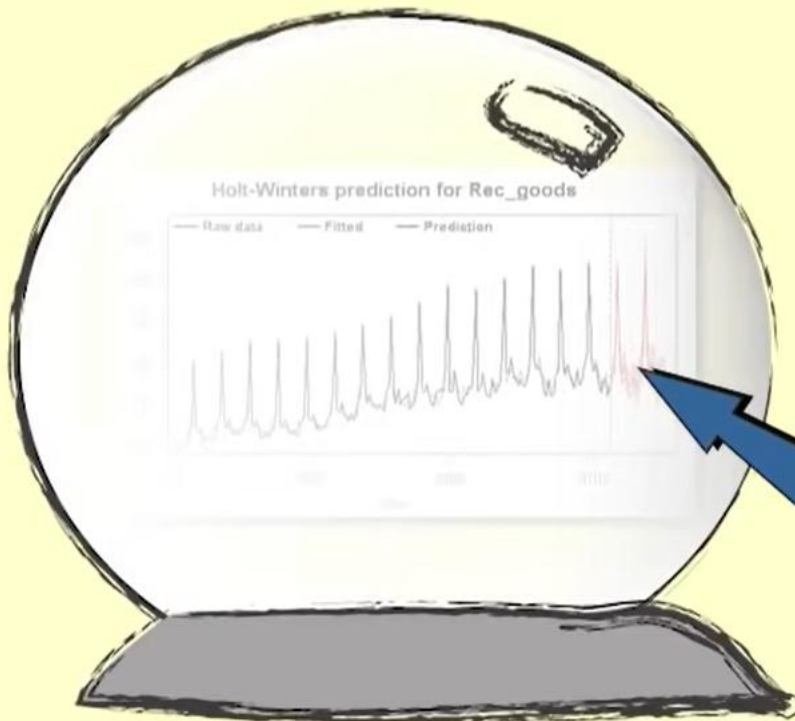
Data





Month	Recreational goods
Jan 2000	134.1
Feb 2000	133
Mar 2000	136.2
Apr 2000	120.4
May 2000	126.2
Jun 2000	119.8
Jul 2000	122.5
Aug 2000	122.3
Sep 2000	131.2
Oct 2000	138.7
Nov 2000	158.4
Dec 2000	232.2
Jan 2001	143.1
Feb 2001	142.4
Mar 2001	152.6
Apr 2001	139.9
May 2001	135.6
Jun 2001	129.4
Jul 2001	138.5
Aug 2001	130.2

Data



Time Series

It is an **arrangement** of the statistical data in accordance with the **time of occurrence**.

Most of the series related to **Economics, Business and Commerce**, e.g., the series related to **prices**, production and consumption of various commodities; **investment, sales and profit of business house**.

Time Series analysis

A time series is a series of figures recorded over **time**, e.g. unemployment over the last 5 **years**, output over the last 12 **months**, etc.

Time series analysis is a technique used to:

- identify whether there is any **underlying historical trend**
- use this analysis of the historical trend to forecast the trend into the future
- identify whether there are any **seasonal variations** around the trend
- apply estimated seasonal variations to a trend line forecast in order to prepare a forecast season by season.

Time Series Forecasting

- Time series forecasting occurs when you make scientific predictions based on historical time stamped data.
- It involves building models through historical analysis and using them to make observations and drive future strategic decision-making.
- An important distinction in forecasting is that at the time of the work, the future outcome is completely unavailable and can only be estimated through careful analysis and evidence-based priors.

Web Traffic Time-series data

10am 11am 12pm 1pm 2pm 3pm

T1 T2 T3 T4 T5 T6

300 310 420 530 640



Applications of time series forecasting

Forecasting has a range of applications in various industries.

weather forecasting,

climate forecasting,

economic forecasting,

healthcare forecasting

engineering forecasting,

finance forecasting,

retail forecasting,

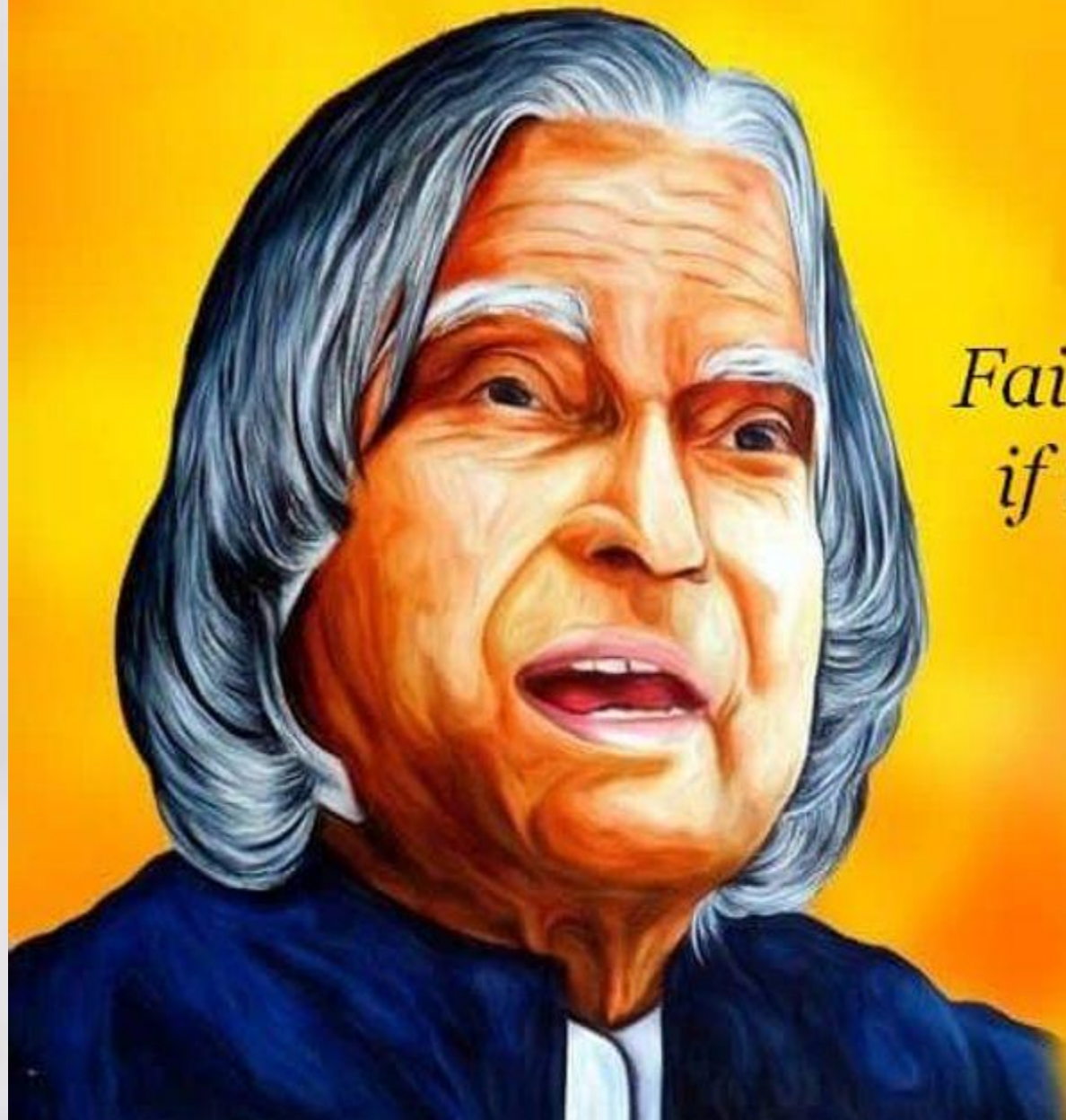
business forecasting,

environmental studies forecasting, social studies forecasting, and more.

Applications of time series forecasting

Forecasting has a range of applications in various industries.

Basically anyone who has consistent historical data can analyze that data with time series analysis methods and then model, forecasting, and predict. For some industries, the entire point of time series analysis is to facilitate forecasting.



*Failure will never overtake me
if my definition to succeed is
strong enough*

Abdul Kalam

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Root Mean Square Error,

Mean Absolute Percentage Error,

Mean Absolute Scaled Error

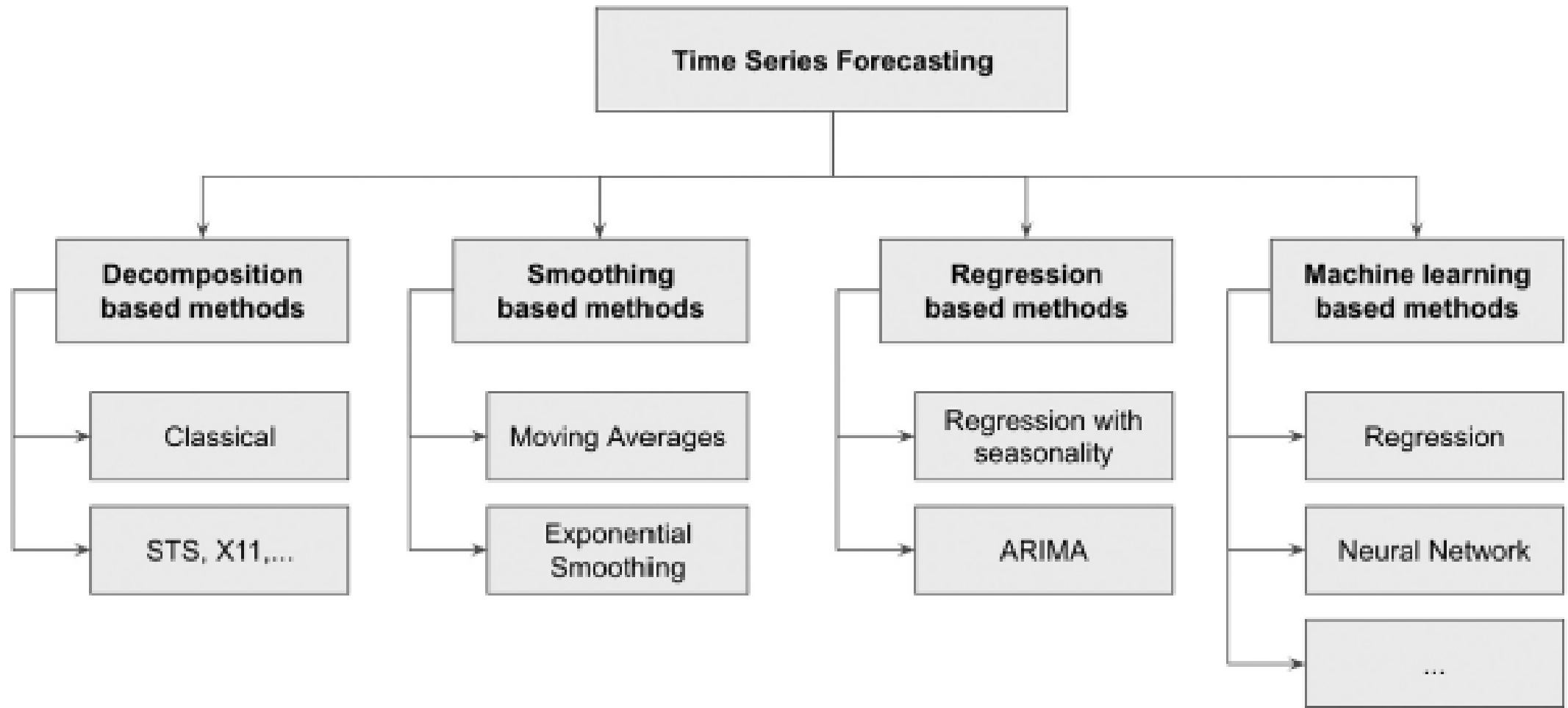
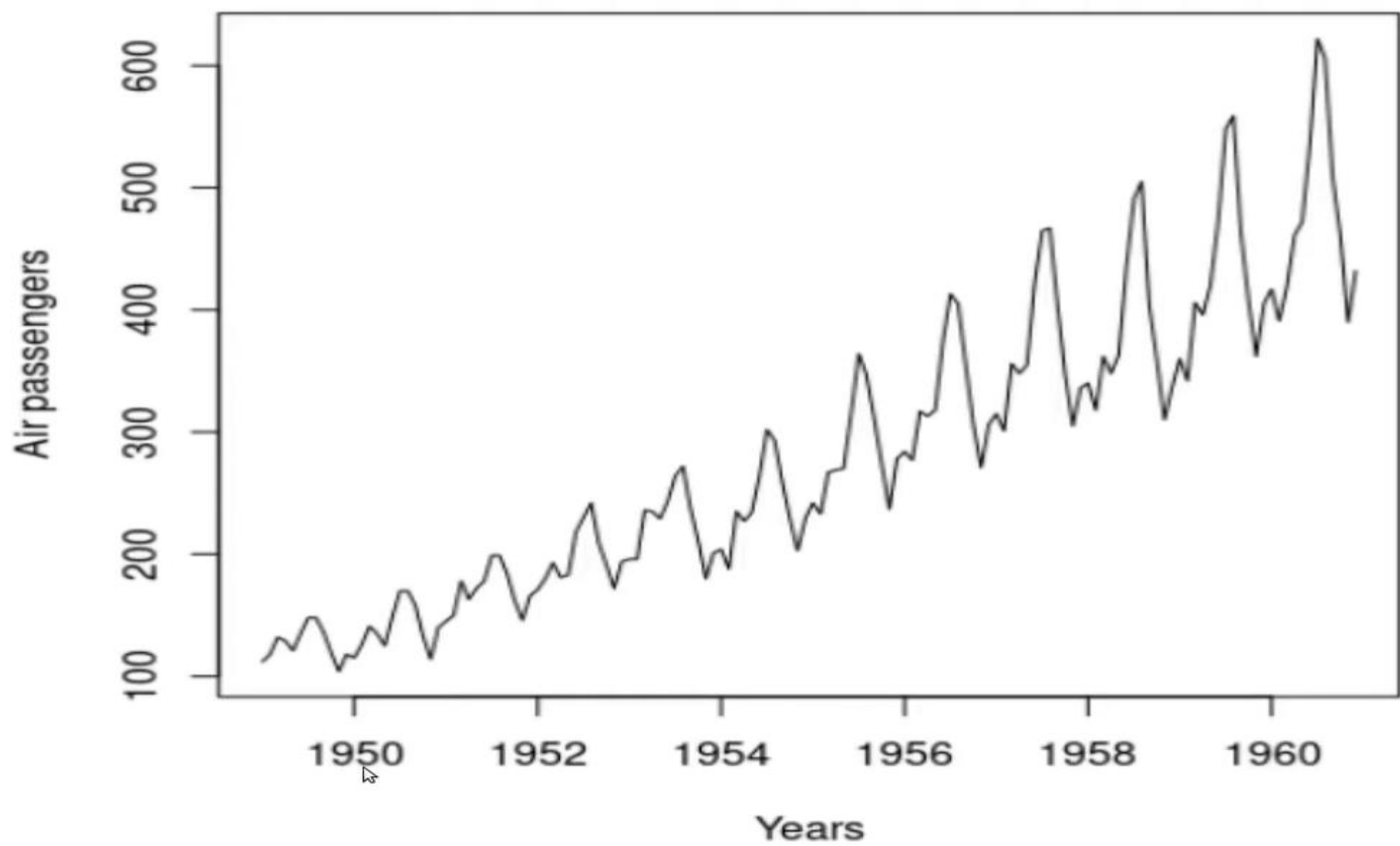


FIGURE 12.3

Taxonomy of time series forecasting techniques.

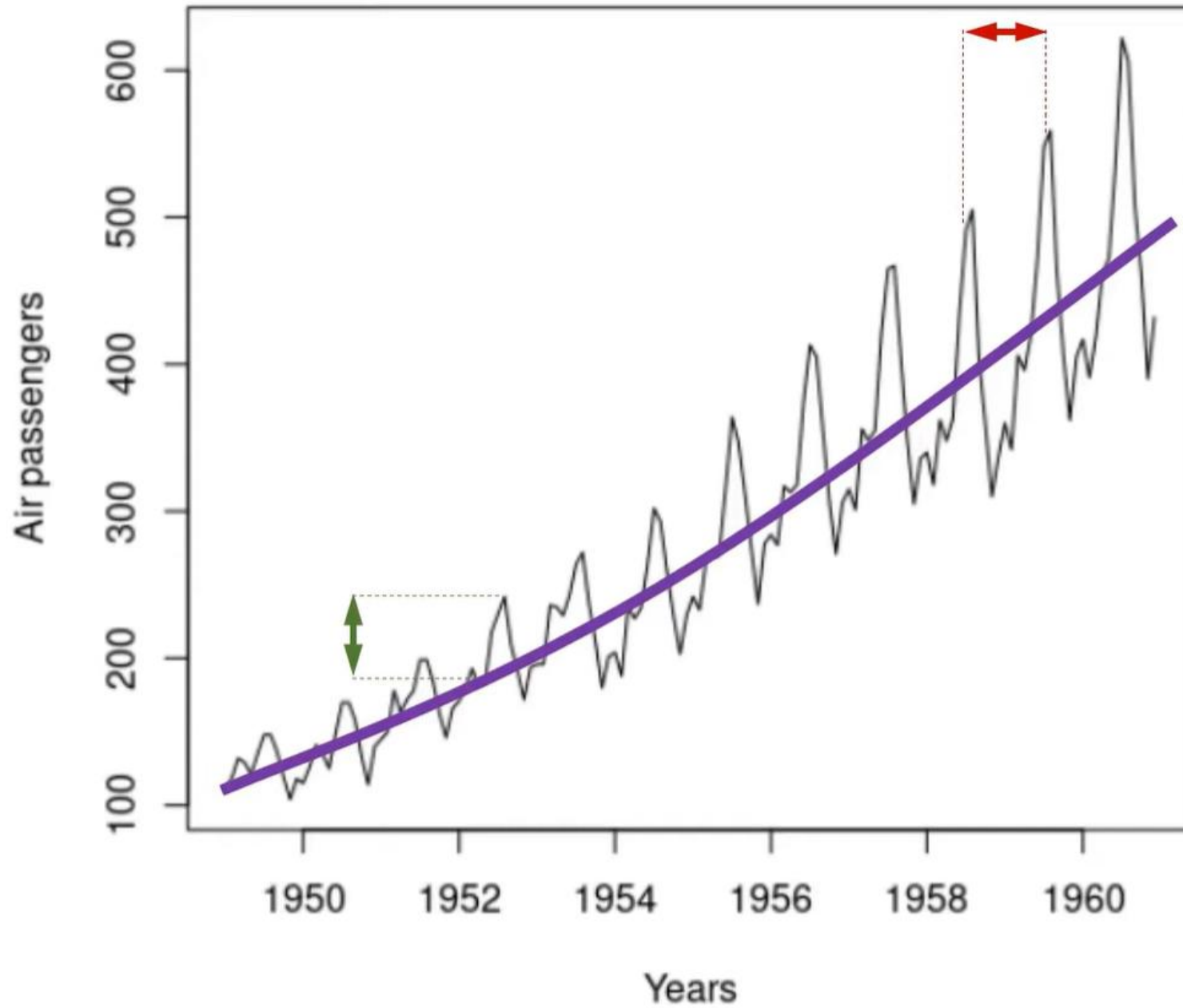


Rolling Statistics and Stationarity in Time-series

A stationary time series is a data that has a constant mean and constant variance.

If I take a mean of T_1 and T_2 and compare it with the mean of T_4 and T_5 then is it the same, and if different, how much difference is there? So, constant mean means this difference should be less, and the same with variance.

If the time series is not stationary, we have to make it stationary and then proceed with modelling. Rolling statistics help us in making time series stationary. so basically rolling statistics calculates moving average. To calculate the moving average we need to define the window size which is basically how much past values to be considered.



4

Some sort of periodicity

Variability grows with time

There is a growing trend

Time series components

Trend

Seasonality

Irregularities

Cycles

Variation

Components of a time series

The pattern or behavior of the data in a time series has several components.

Theoretically, any time series can be decomposed into:

Secular Trend

Cyclical variation

Seasonal variation

Irregular

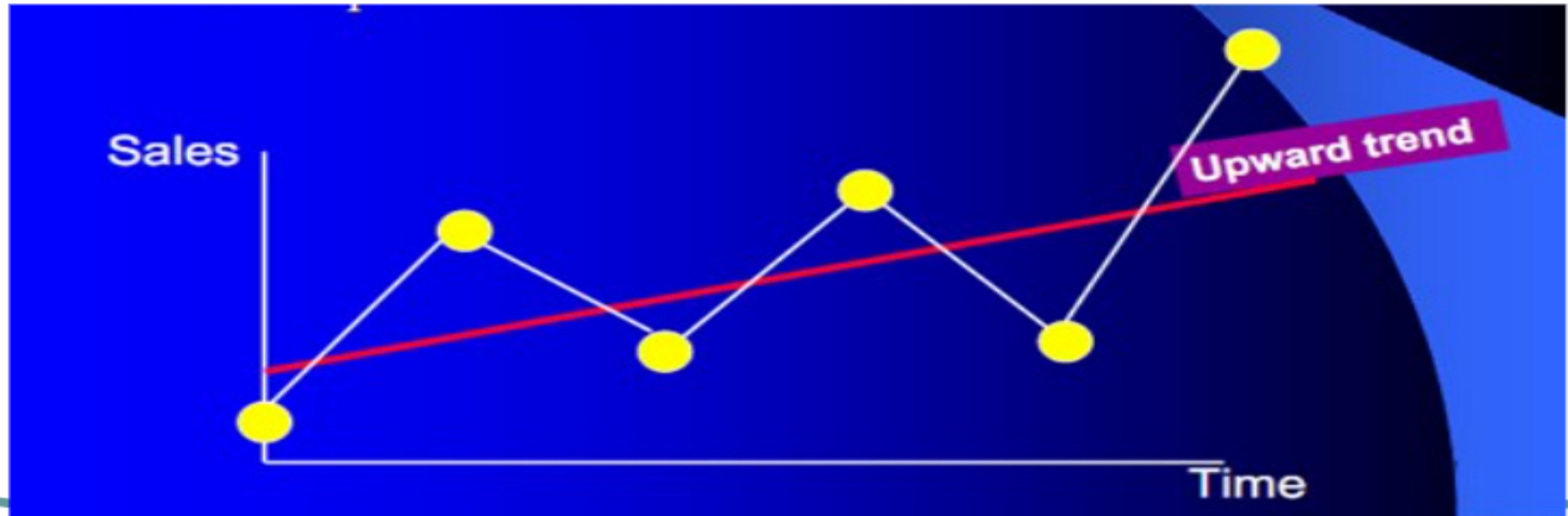
However, this decomposition is often not straight-forward because these factors interact.

Trend



Trend component

- The *trend component* accounts for the gradual shifting of the time series to relatively higher or lower values over a *long period* of time.
- Trend is usually the result of long-term factors such as changes in the population, demographics, technology, or consumer preferences.



The trend

Most time series follow some sort of long term movement. In time series analysis the trend is measured by:

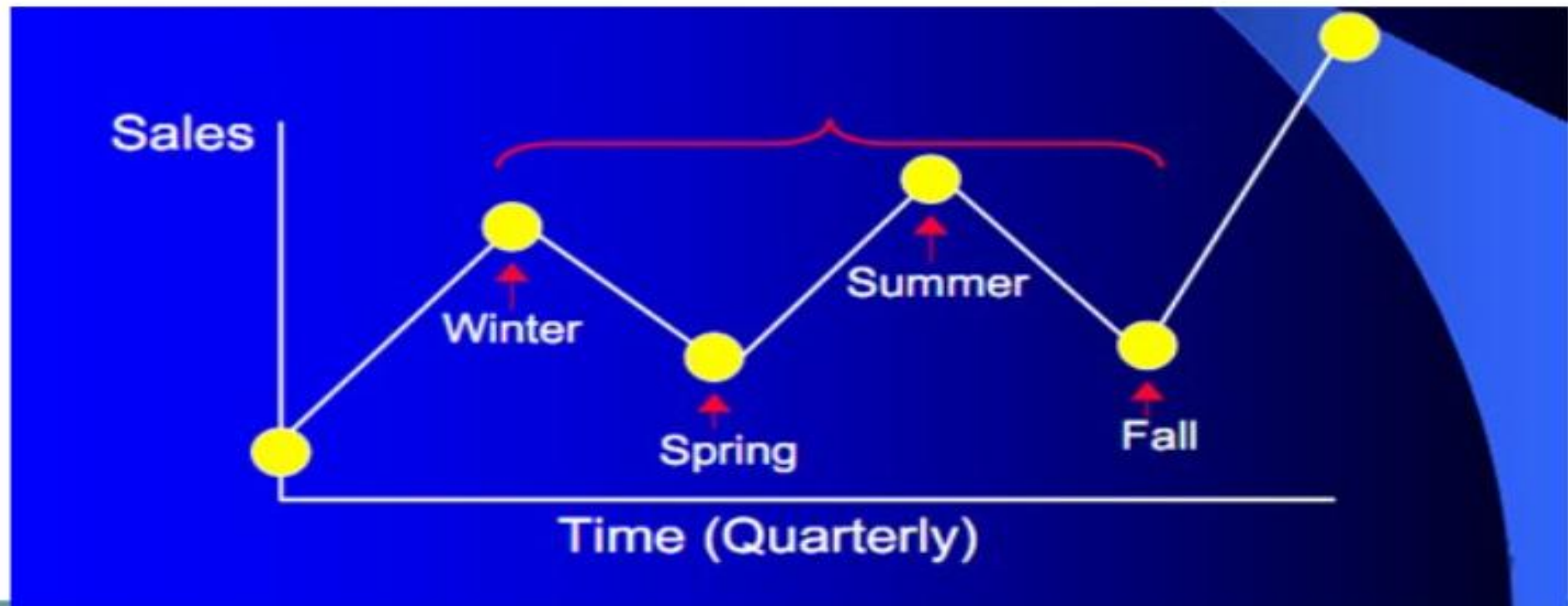
- (1) *Inspection*. A graph of the data is produced and the trend line is drawn by eye with the aim of plotting the line so that it lies in the middle of the data points.
- (2) *Least squares regression analysis*. x represents time (each month would be given a number e.g. January =1, February =2 etc) and y is the data.
- (3) *Moving averages*. This method attempts to remove seasonal or cyclical variations by a process of averaging.

Seasonality



Seasonal component

- The seasonal component accounts for regular patterns of variability within certain time periods, such as a year.
- The variability does not always correspond with the seasons of the year (i.e. winter, spring, summer, fall).
- There can be, for example, within-week or within-day “seasonal” behavior.

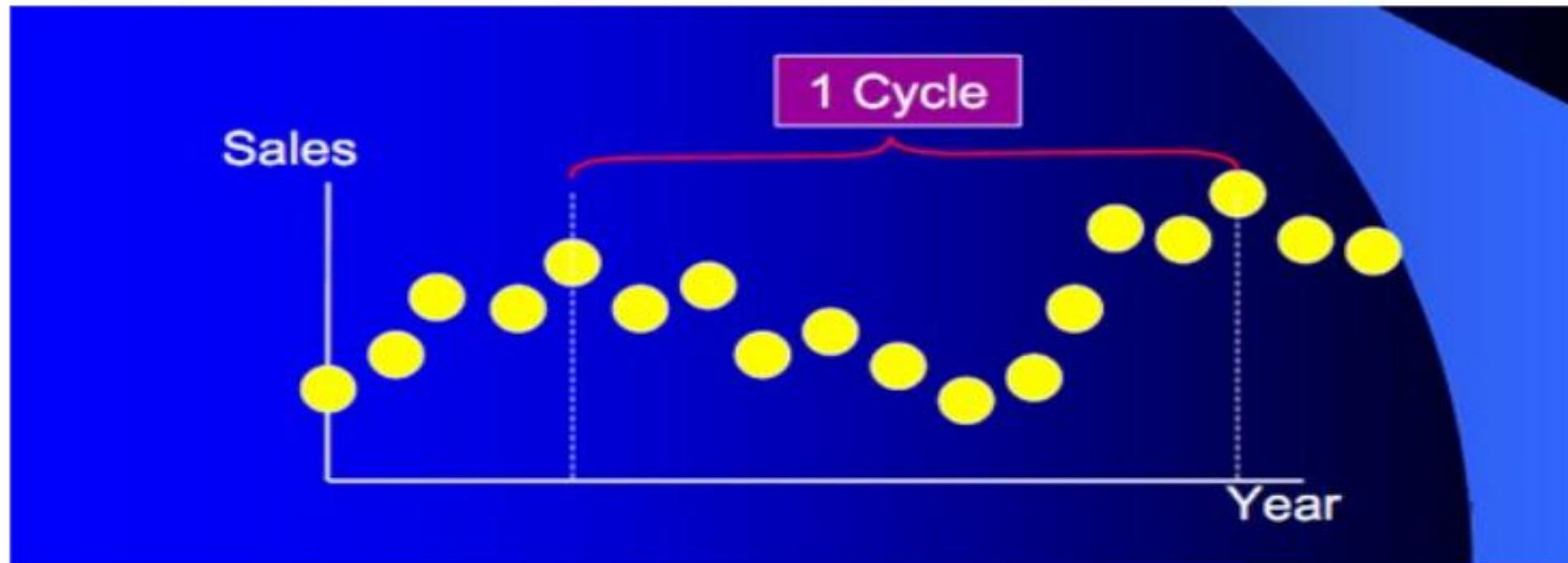


Cycles

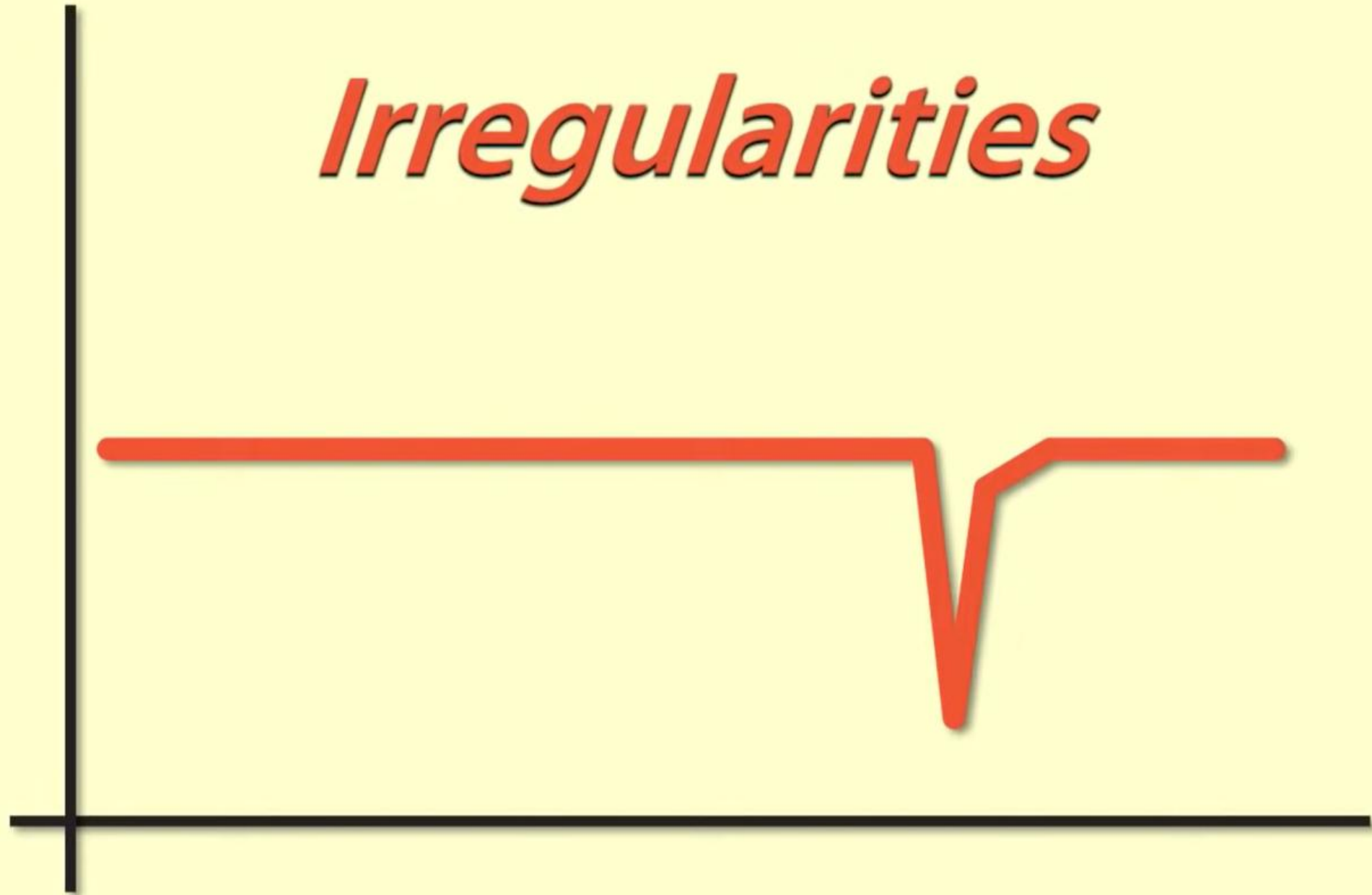


Cyclical component

- Any regular pattern of sequences of values above and below the trend line lasting more than one year can be attributed to the *cyclical component*.
- Usually, this component is due to multiyear cyclical movements in the economy.



Irregularities



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Mean Absolute Scaled Error

Decomposition methods

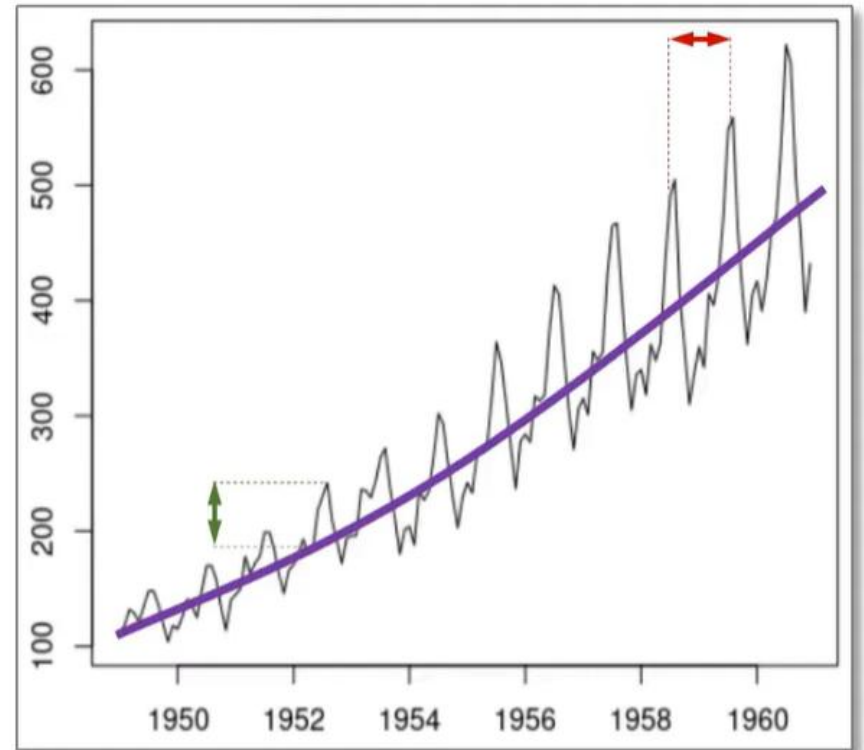
$$y(t) = f(\tau(t), \sigma(t), \varepsilon(t))$$

$y(t)$: time series value at time t

$\tau(t)$: **trend cycle** component at time t

$\sigma(t)$: **seasonal** component at time t

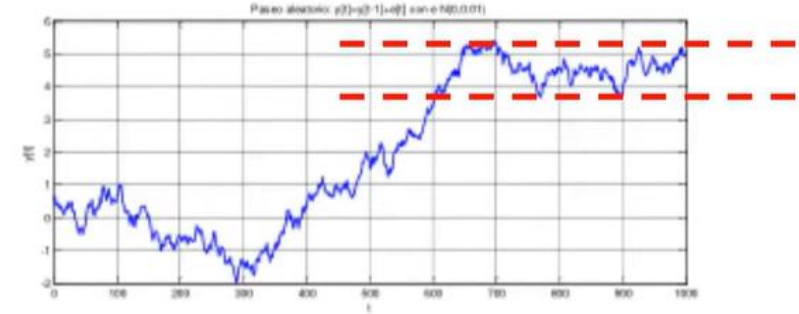
$\varepsilon(t)$: **irregular** (or remainder) component at time t



Types of decomposition

Additive Model:

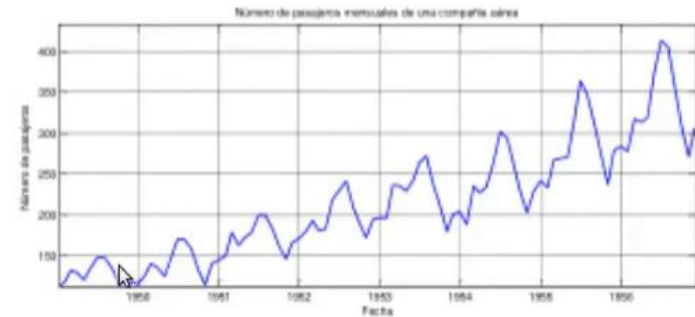
$$y(t) = \tau(t) + \sigma(t) + \varepsilon(t)$$



The additive model is appropriate if the magnitude of the seasonal fluctuations does not vary with the level of the time series

Multiplicative Model:

$$y(t) = \tau(t) \times \sigma(t) \times \varepsilon(t)$$



Multiplicative decomposition is more prevalent with economic series because most seasonal economic series do have seasonal variation which increases with the level of the series.

Additive and multiplicative Time-series

Additive Time Series

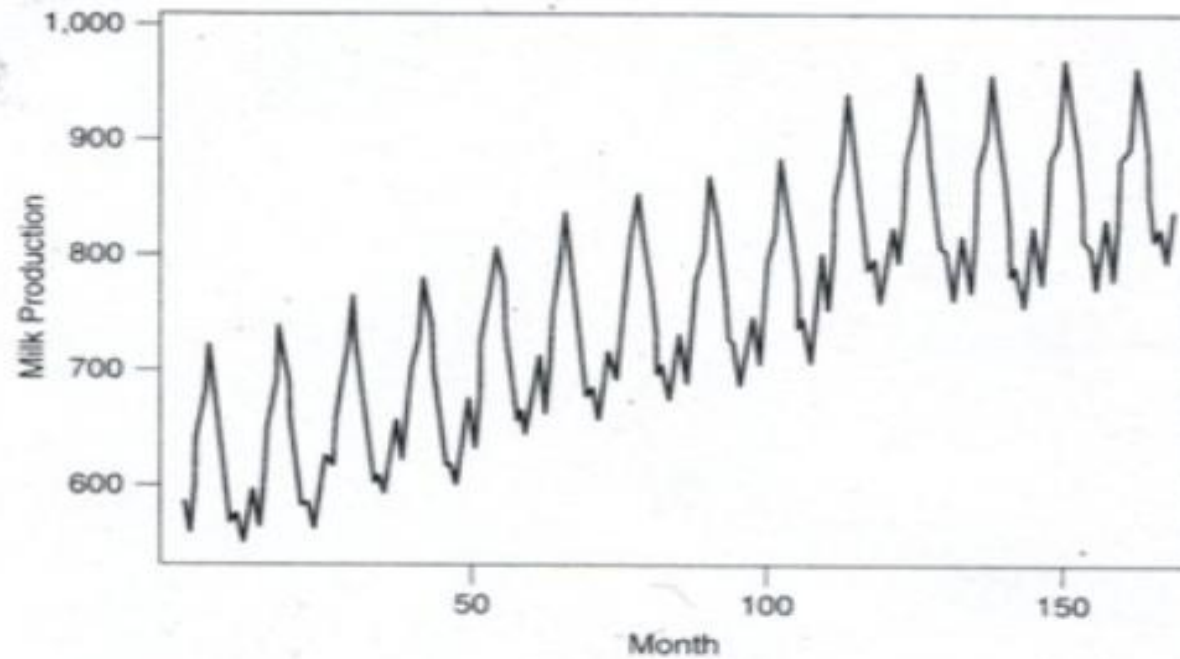
Value = Base Level + Trend + Seasonality + Error

Multiplicative Time Series

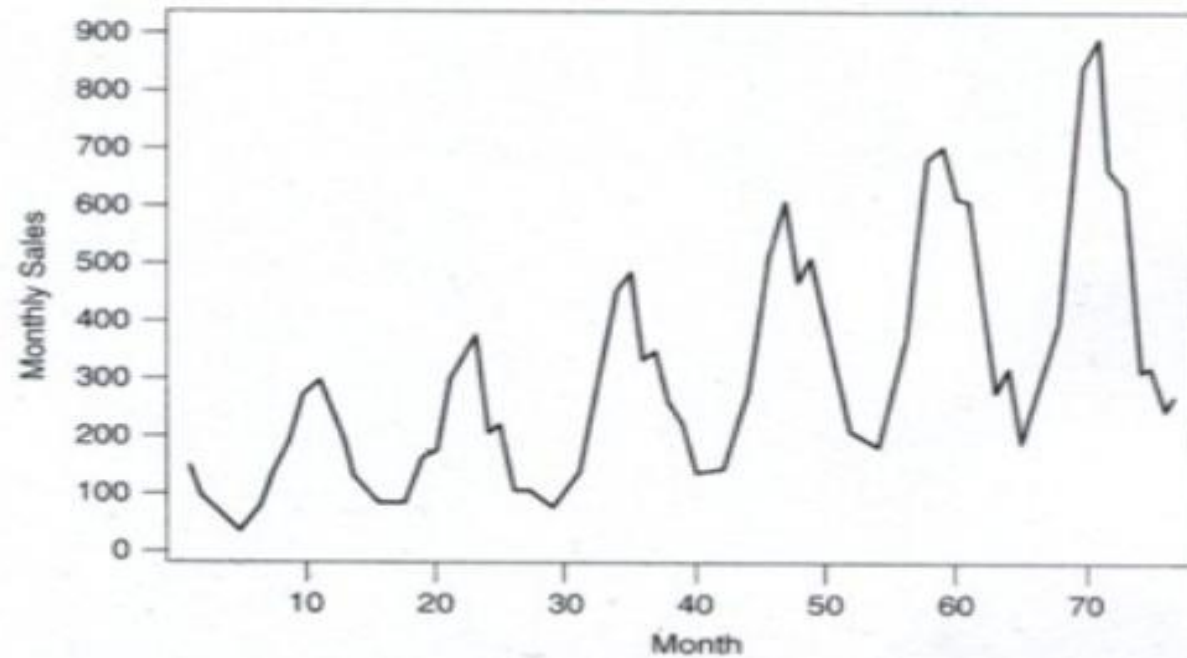
Value = Base Level * Trend * Seasonality * Error

Additive and multiplicative models

- The additive model works best when the time series has roughly the same variability through the length of the series.
 - That is, all the values of the series fall within a band with constant width centered on the trend.
- The multiplicative model works best when the variability of the time series increased with the level.
 - That is the values of the series become larger as the trend increases.
- See the figure in the next slide.
- Most economic time series have seasonal variation that increases with the level of the series. So *multiplicative model* is suitable to them.



(a) A time series with constant variability



(b) A time series with variability increasing with level

12.1.1 Classical Decomposition

The classical decomposition technique

- first estimates the trend component by calculating the long term (say 12 month) moving average.
- The trend component is removed from the time series to get remaining seasonal and noise components.
- The seasonal component can be estimated by average Jan, Feb, Mar, . . . , variance in the remaining series.
- Once the trend and seasonal components are removed, what is left is noise.

12.1.1 Classical Decomposition

Algorithm

1. Estimate the trend T_t : If m is even, calculate $2 \times m$ moving average (m -MA and then a 2-MA); if m is odd, calculate m -moving average. A moving average is the average of last m data points.
2. Calculate detrended series: Calculate $y_t - T_t$ for each data point in the series.
3. Estimate the seasonal component S_t : average $(y_t - T_t)$ for each m period. For example, calculate the average of all January values of $(y_t - T_t)$ and repeat for all the months. Normalize seasonal value in such a way that mean is zero.
4. Calculate the noise component E_t : $E_t = (y_t - T_t - S_t)$ for each data point in the series.

Multiplicative decomposition is similar to additive decomposition: Replace subtraction with division in the algorithm

Forecasting Using Decomposed Data

While decomposing the time series is used to describe and increase the understanding of time series data, it is useful for forecasting as well. The idea is to breakdown the time series into its parts, forecast the parts, and put it back together for forecasted future time series values.

$$\hat{y}_t = \hat{S}_t + \hat{T}_t$$

1. Step 1:
 1. If m is an even number, compute the trend-cycle component using a $2 \times m$ -MA.
 2. If m is an odd number, compute the trend-cycle component using an m -MA.
2. Step 2: Calculate the detrended series:
3. Step 3: Average the detrended values for that season to estimate the seasonal component. For monthly data, the seasonal component for March is the average of all the detrended March values in the data.
4. Step 4: The remainder component is calculated

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Methods of smoothing time series

- Arithmetic Moving Average
- Exponential Smoothing Methods
- Holt-Winters method for Exponential Smoothing

Smoothing a time series: to eliminate some of short-term fluctuations.

Smoothing also can be done to remove seasonal fluctuations, i.e., to *deseasonalize* a time series.

These models are *deterministic* in that no reference is made to the sources or nature of the underlying randomness in the series.

The models involves *extrapolation* techniques.

Averaging Methods

- Simple Averages - quick, inexpensive (should only be used on stationary data)
- *Moving Average* method consists of computing an average of the most recent n data values for the series and using this average for forecasting the value of the time series for the next period.
- Moving averages are useful if one can assume item to be forecast will stay steady over time.
- Series of arithmetic means – used only for smoothing, provides overall impression of data over time

$$\text{Moving Average} = \frac{\sum (\text{most recent } n \text{ data items})}{n}$$

Moving average methods

- Works best with stationary data.
- The smaller the number, the more weight given to recent periods.
- A smaller number is desirable when there are sudden shifts in the level of the series.
- The greater the number, less weight is given to more recent periods.
- The larger the order of the moving average, the greater the smoothing effect. Larger n when there are wide, infrequent *fluctuations* in the data.
- By smoothing recent actual values, removes randomness.

12.2 SMOOTHING BASED METHODS

In the smoothing based approaches, an observation is a function of past few observations. It is helpful to start out with a basic notation system for time series in order to understand the different smoothing methodologies.

- *Time periods:* $t = 1, 2, 3, \dots, n$. Time periods can be seconds, days, weeks, months, or years depending on the problem.
- *Data series:* Observation corresponding to each time period above: $y_1, y_2, y_3, \dots, y_n$.
- *Forecasts:* F_{n+h} is the forecast for the h^{th} time period following n . Usually $h = 1$, the next time period following the last data point. However h can be greater than 1. h is called the *horizon*.
- *Forecast errors:* $e_t = y_t - F_t$ for any given time, t .

In order to explain the different methods, a simple time series data function will be used, $Y(t)$. Y is the *observed* value of the time series at any time t . Furthermore, the observations are made at a constant time interval. If in the case of intermittent data, one can assume that an interpolation scheme is applied to obtain equally spaced (in time) data points.

12.2.1 Simple Forecasting Methods

Naïve Method

Probably the simplest forecasting “model.” Here one simply assumes that F_{n+1} , the forecast for the next period in the series, is given by the last data point of the series, y_n

$$F_{n+1} = y_n \quad (12.5)$$

Seasonal Naïve Method

If the time series is seasonal, one can forecast better than the naive point estimate. The forecast can assume the value as the previous value of the same season. For example, the next January revenue can be assumed as the last known January revenue.

$$F_{n+1} = y_{n-s} \quad (12.6)$$

where s is the seasonal period. In the case of monthly data with yearly seasonality, seasonal period is 12.

Average Method

Moving up a level, one could compute the next data point as an average of all the data points in the series. In other words, this model calculates the forecasted value, F_{n+1} , as:

$$F_{n+1} = \text{Average}(y_n, y_{n-1}, y_{n-2}, \dots, y_1) \quad (12.7)$$

Suppose one has monthly data from January 2010 to December 2010 and they want to predict the next January 2011 value, they would simply average the values from January 2010 to December 2010.

Moving Average Smoothing

select a window of the last “k” periods to calculate the average, and as the actual data grows over time, one can always take the last k samples to average, that is,

$$n, n - 1, \dots, n - k + 1$$

Suppose in the simple example of the window $k = 3$; then to predict the January 2021 data, a three-month average would be taken using the last three months. When the actual data from January comes in, the February 2021 value is forecasted using January 2021 (n), December 2020 ($n - 1$) and November 2020 ($n - 3 + 1$ or $n - 2$).

Moving average smoothens the seasonal information in the time series.

$$F_{n+1} = \text{Average}(y_n, y_{n-1}, \dots, y_{n-k})$$

Weighted Moving Average Smoothing

For some cases, the most recent value could have more influence than some of the earlier values. Most exponential growth occurs due to this simple effect. The forecast for the next period is given by the model:

$$F_{n+1} = (a \times y_n + b \times y_{n-1} + c \times y_{n-k}) / (a + b + c) \quad \text{where typically } a > b > c. \quad (12.9)$$

Exponential Smoothing

Exponential smoothing is the weighted average of the past data, with the recent data points given more weight than earlier data points. The weights decay exponentially towards the earlier data points, hence, the name.

To forecast the future values using exponential smoothing, be rewritten as:

$$F_{n+1} = \alpha \times y_n + (1 - \alpha) \times F_n$$

A higher α value gives an exact fit and a lower value gives a smoother fit. Going back to the monthly example, if one wanted to make the February 2011 forecast using not only the actual January 2011 value but also the previously *forecasted* January 2011 value, the new forecast would have “learnt” the data better. This is the concept behind basic exponential smoothing (Brown, 1956).

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REGRESSION BASED METHODS

In the regression based methods, the variable *time* is the predictor or independent variable and the time series value is the dependent variable. Regression based methods are generally preferable when the time series appears to have a global pattern. The idea is that the model parameters will be able to capture these patterns and, thus, enable one to make predictions for any step ahead in the future under the assumption that this pattern is going to be repeated. For a time series with local patterns instead of a global pattern, using a regression based approach requires one to specify how and when the patterns change, which is difficult. For such a series, smoothing approaches work best because these methods usually rely on extrapolating the most recent local pattern as seen earlier.

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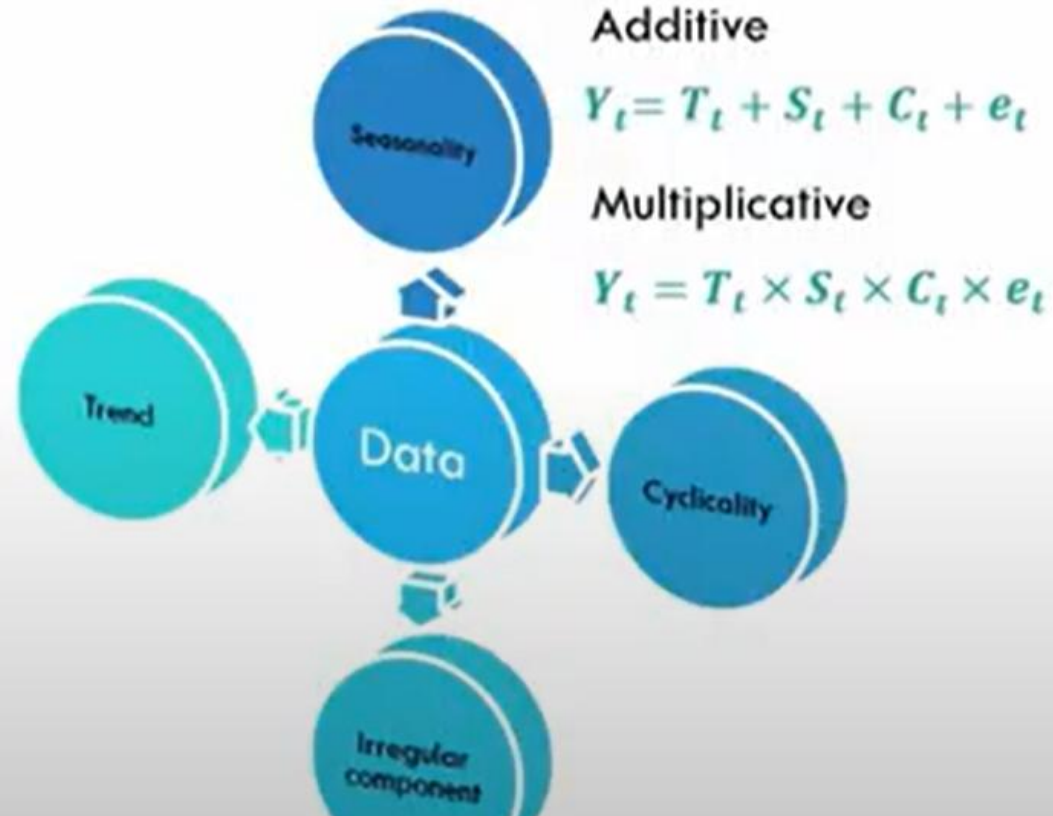
Mean Absolute Percentage Error,

Mean Absolute Scaled Error

ARIMA

PARADIGM SHIFT

Time Series

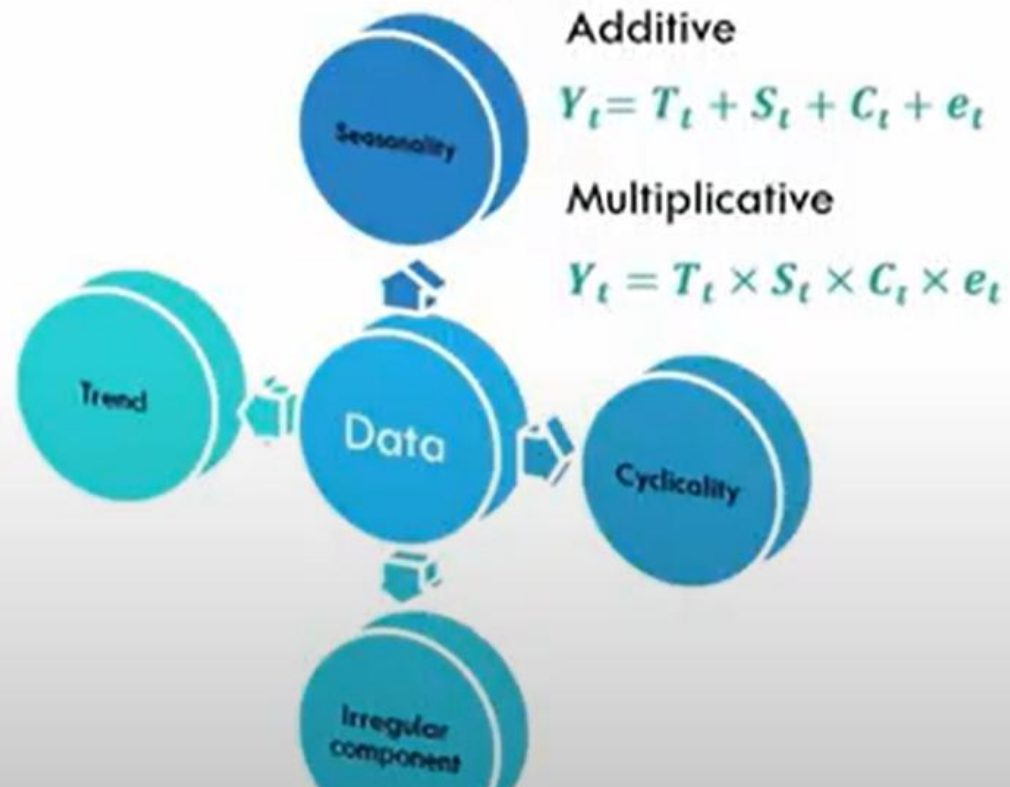


Box-Jenkins Method

ARIMA

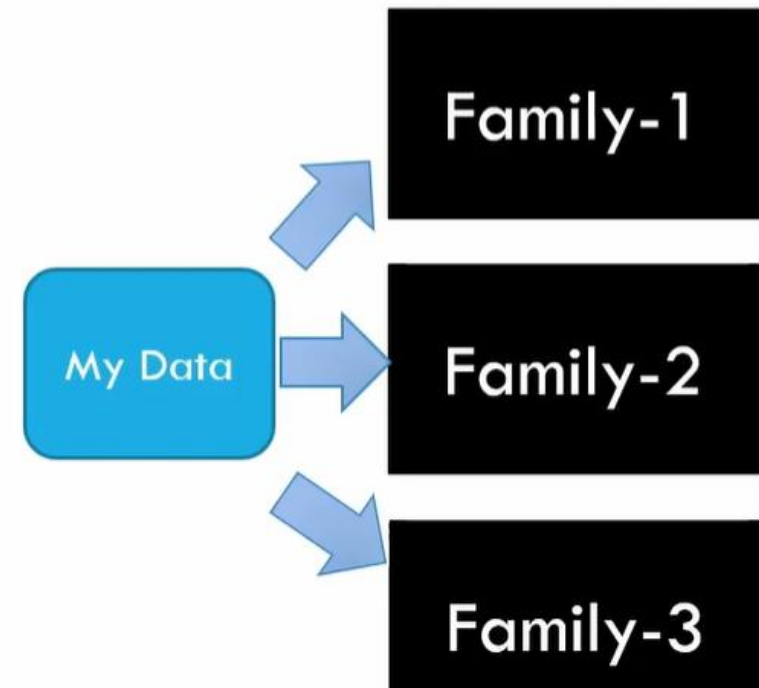
PARADIGM SHIFT

Time Series



Box-Jenkins Method

Any data is part of a family



ARIMA

Box-Jenkins Method

An iterative approach consists of 3 steps

Identification



Estimation



Diagnostic
Checking

ARIMA

Components of ARIMA

Three components of ARIMA – AR, I and MA



Components of ARIMA

Three components of ARIMA – AR, I and MA

AR

Auto regressive part which accounts for series dependence on its past behavior

p

$p = 0/1/2/.../p$: Observations depend on 'p' previous values

Components of ARIMA

Three components of ARIMA – AR, I and MA

AR

Auto regressive part which accounts for series dependence on its past behavior

p

$p = 0/1/2/.../p$: Observations depend on 'p' previous values

AUTO-REGRESSIVE (AR) MODELS

Period	Actual	Forecast	Error
1	Y1	F1	$\varepsilon_1 = A_1 - F_1$
2	Y2	F2	$\varepsilon_2 = A_2 - F_2$
3	Y3	F3	$\varepsilon_3 = A_3 - F_3$

Components of ARIMA

Three components of ARIMA – AR, I and MA

AUTO-REGRESSIVE (AR) MODELS

Period	Actual	Forecast	Error
1	Y1	F1	$\varepsilon_1 = A_1 - F_1$
2	Y2	F2	$\varepsilon_2 = A_2 - F_2$
3	Y3	F3	$\varepsilon_3 = A_3 - F_3$

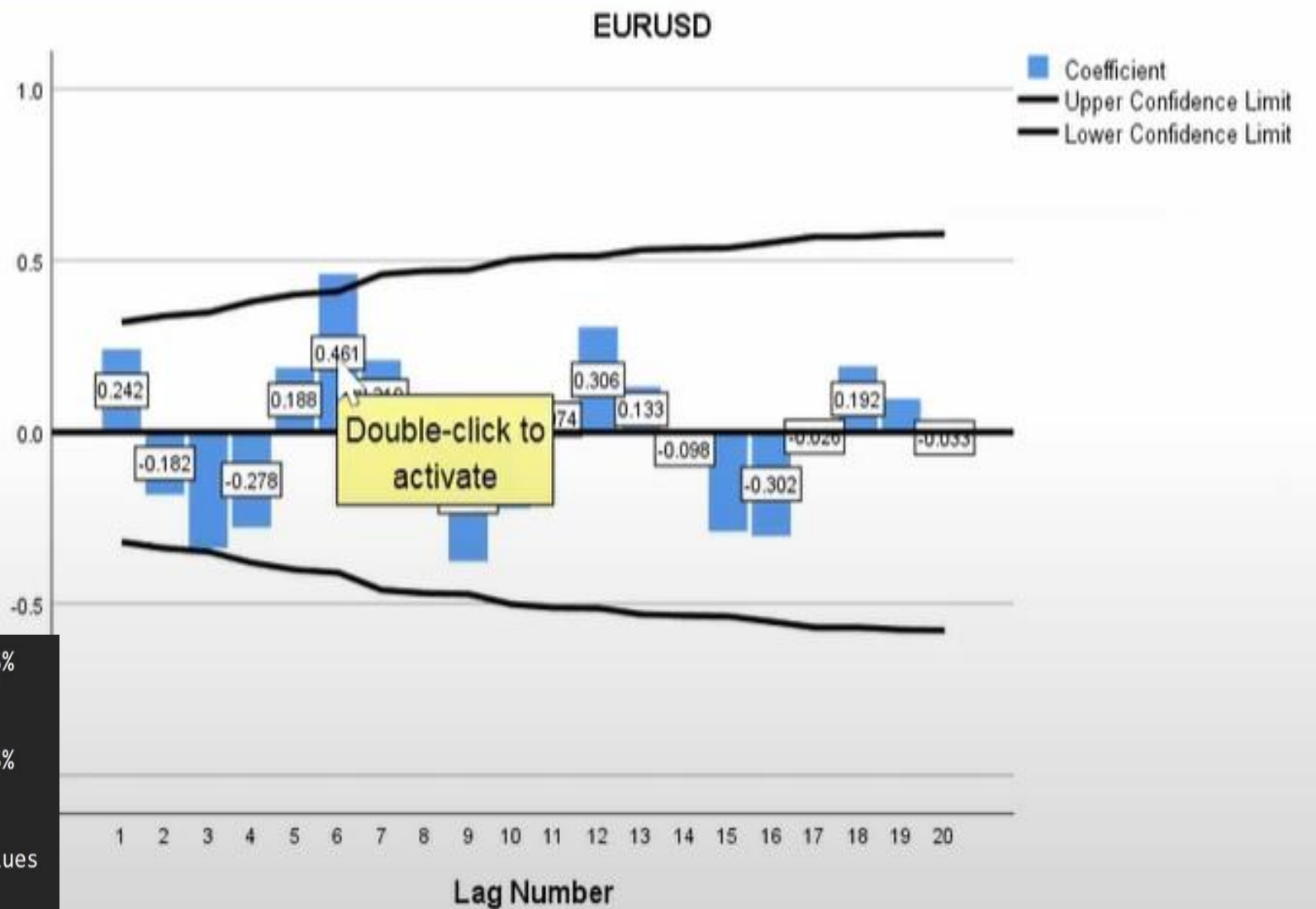
$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \varepsilon_t$$

AR(p) model

Day	EURUSD	Lag 1	Lag 2	Lag 3	Lag 4					
1	1.05078									
2	1.05541	1.05078								
3	1.01833	1.05541	1.05078							
4	1.09147	1.01833	1.05541	1.05078				<u>Autocorrelation</u>		
5	1.08753	1.09147	1.01833	1.05541	1.05078			Correlation(Data, Lagged data)		
6	1.0511	1.08753	1.09147	1.01833	1.05541					
7	1.05219	1.0511	1.08753	1.09147	1.01833					
8	1.05663	1.05219	1.0511	1.08753	1.09147					
9	1.05002	1.05663	1.05219	1.0511	1.08753					
36	1.05627	1.08118	1.0526	1.00219	1.00404					
37	1.13485	1.05627	1.08118	1.0526	1.00219					
38	1.05494	1.13485	1.05627	1.08118	1.0526					
39	1.04963	1.05494	1.13485	1.05627	1.08118					

Day	EURUSD	Lag 1	L
1	1.05078		
2	1.05541	1.05078	
3	1.01833	1.05541	1.0
4	1.09147	1.01833	1.0
5	1.08753	1.09147	1.0
6	1.0511	1.08753	1.0
7	1.05219	1.0511	1.0
8	1.05663	1.05219	1.0
9	1.05002	1.05663	1.0

ACF



1. If any of the autocorrelation values are outside the 95% confidence level ---> There is a trend

2. If none of the autocorrelation values is outside the 95% confidence level ---> Go to step 3

3. Check the LBQ Significance values (P-value). If the values are less than 0.05 ---> There is a trend

If all the values are greater than 0.05 ---> The data is stationary (no trend)

4. For n-season seasonality, there will be a significant autocorrelation value for the n-th lag

Determine of AR term (p)

- Identify using **Partial Autocorrelation (PACF)** plot.
- PACF conveys the pure correlation between a lag and the series.

$$Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + \alpha_3 Y_{t-3}$$

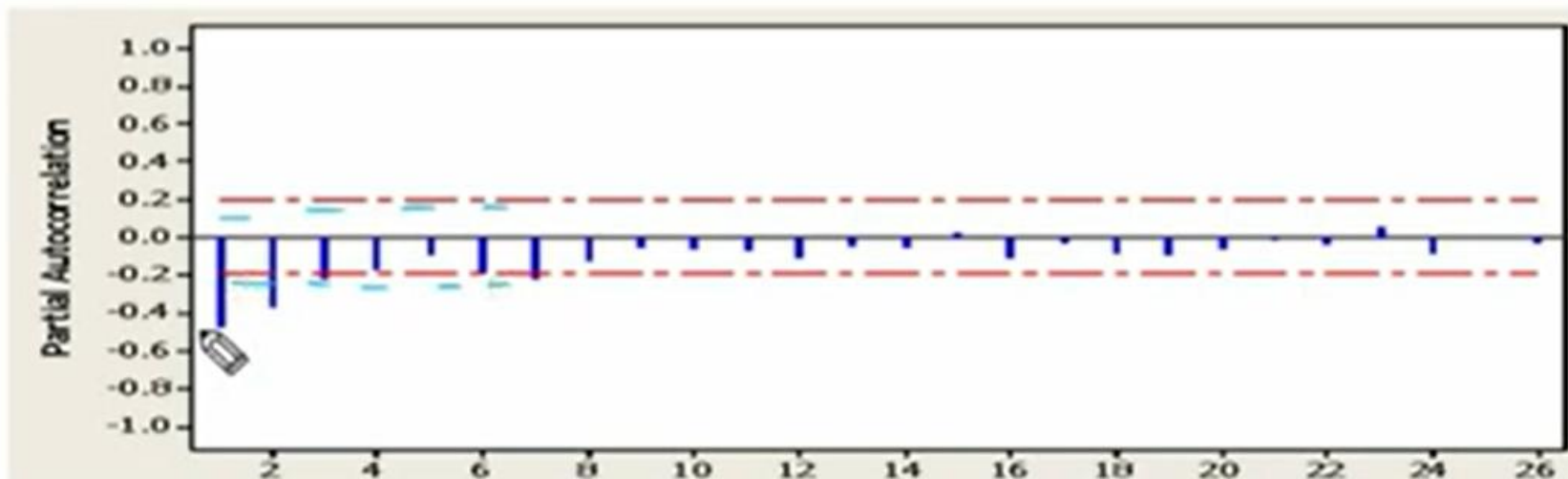
current series

lag 1 of Y

lag 2 of Y

lag 2 of Y

- Take the order of AR term to be equal to as many lags that crosses the significance limit in the PACF plot.



- Identify using **Partial Autocorrelation (PACF)** plot.
- PACF conveys the pure correlation between a lag and the series.

$$Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + \alpha_3 Y_{t-3}$$

current series

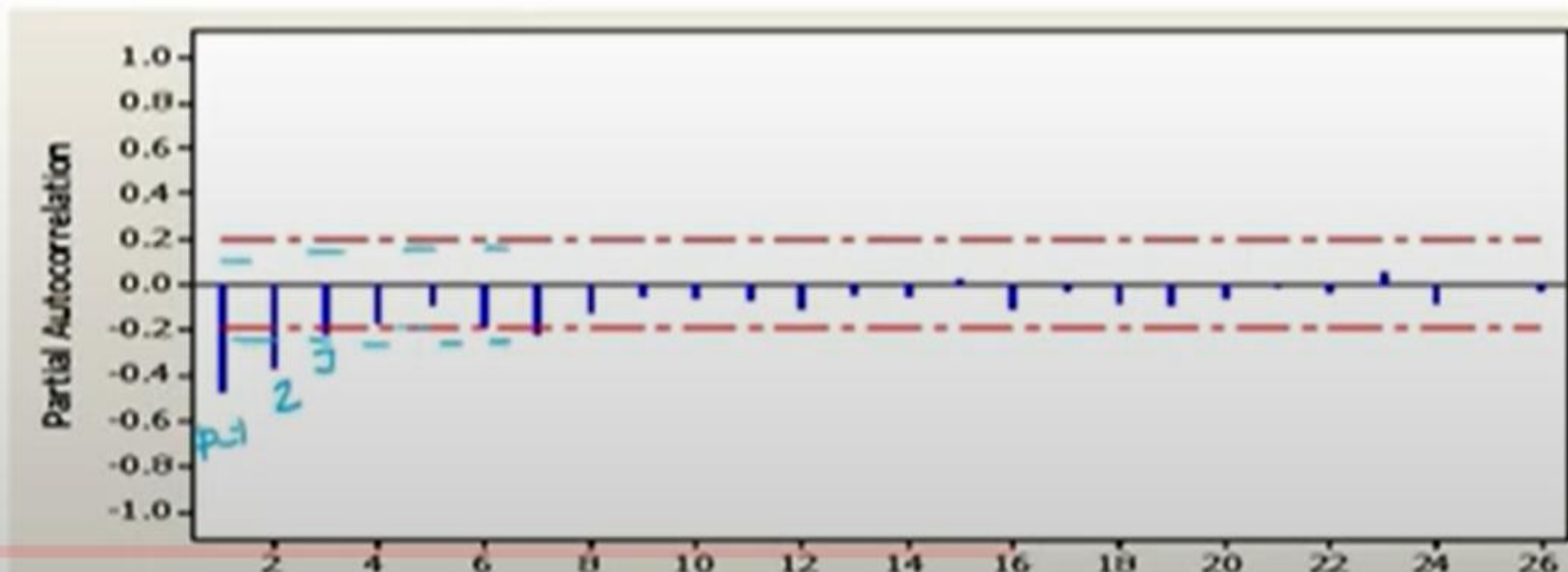
lag 1 of Y

lag 2 of Y

lag 2 of Y

- Take the order of AR term to be equal to as many lags that crosses the significance limit in the PACF plot.

p 1, 2, 3

Components of ARIMA

Three components of ARIMA – AR, I and MA

I

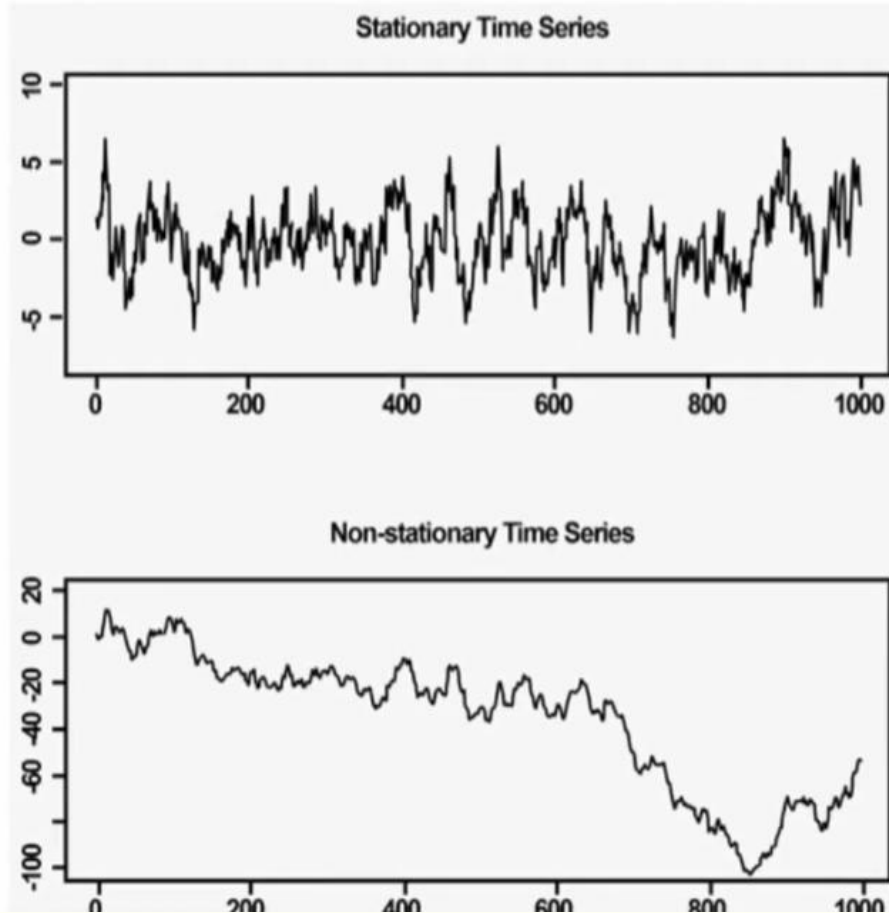
Integrated part which
accounts for the trend
component of a time series

d

- $d = 0$: Series has no trend and is stationary
- $d = 1$: Trend is linear and is differenced once to make it stationary
- $d = 2$: Trend is parabolic and is differenced twice to make it

Stationary

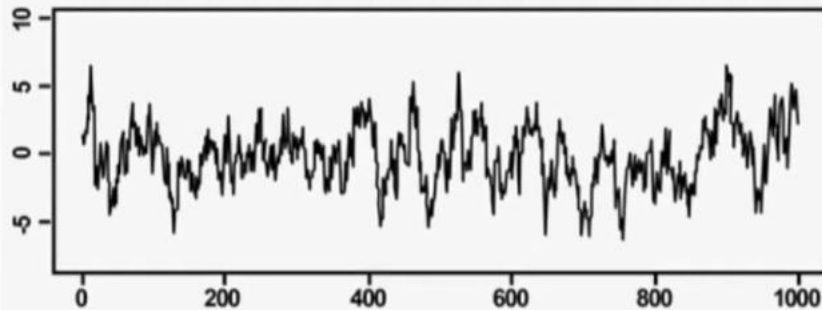
If a time series is stationary, its statistical properties like mean, variance, and covariance will be the same throughout the series, irrespective of the time at which you observe them and **do not follow any trend**.



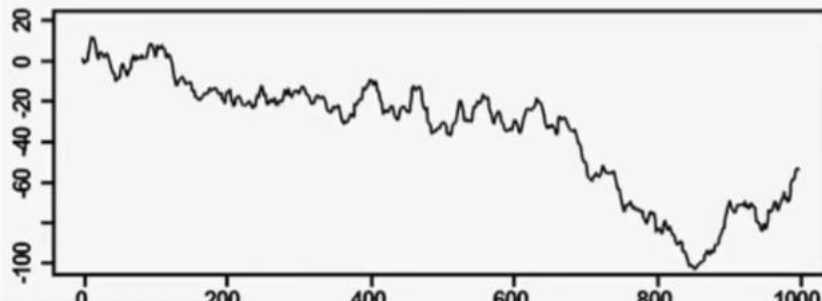
Stationary

If a time series is stationary, its statistical properties like mean, variance, and covariance will be the same throughout the series, irrespective of the time at which you observe them and **do not follow any trend**.

Stationary Time Series



Non-stationary Time Series



Identifying Non-Stationary series:

- Unit root tests (E.g.: Dickey-Fuller test).
- Studying ACF plot: For a stationary time series, the ACF will drop to zero relatively quickly, while the ACF of non-stationary data decreases slowly.

Steps to make time-series stationary:

1. Differencing: Compute the differences between consecutive observations.
2. Transformations (Taking log).

Differencing: Differencing (Make the Series Stationary)

- Compute the differences between consecutive observations.
- It's just like differencing a sloped line so as to make the slope zero.
- Differencing stabilizes the mean of a time series by removing changes in the level of a time series and therefore eliminating (or reducing) trend and seasonality.

For Example:

Original time series	After 1st order differencing	After 2nd order differencing
50		
180	130	
420	240	110
770	350	110
1240	470	120
1820	580	110



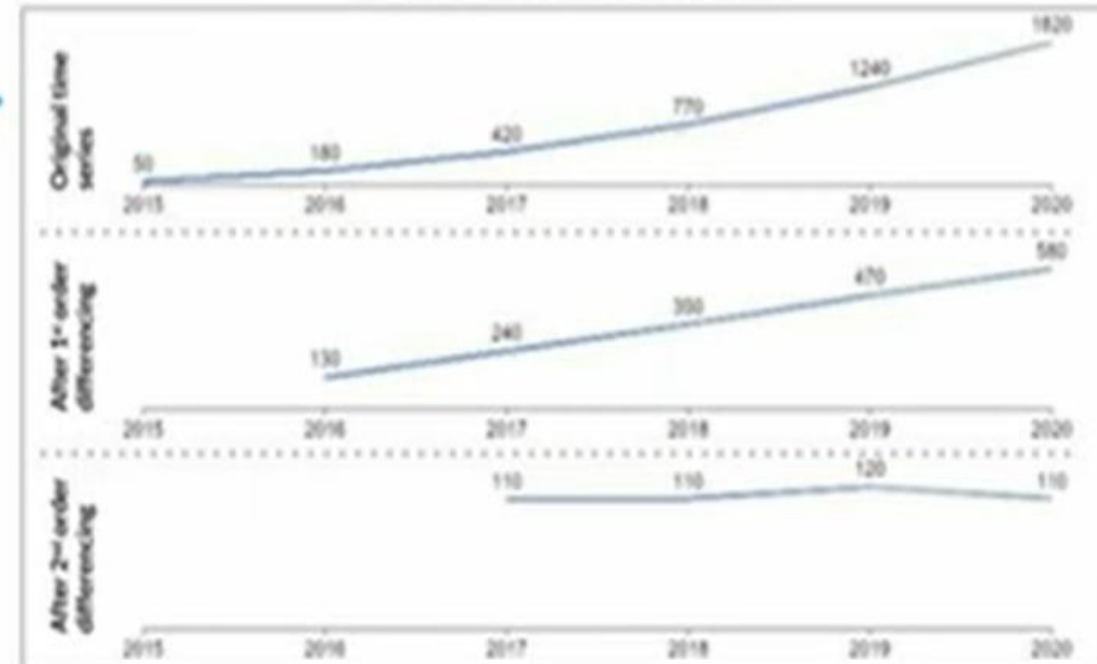
Differencing: Differencing (Make the Series Stationary)

- Compute the differences between consecutive observations.
- It's just like differencing a sloped line so as to make the slope zero.
- Differencing stabilizes the mean of a time series by removing changes in the level of a time series and therefore eliminating (or reducing) trend and seasonality.

For Example:

Original time series	After 1st order differencing	After 2nd order differencing
50		
180	130	
420	240	110
770	350	110
1240	470	120
1820	580	110

DIFFERENCING

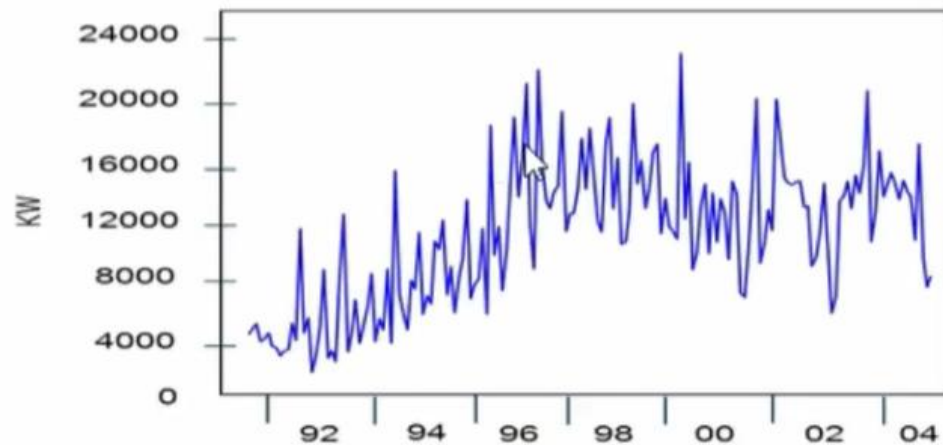


Determine the value of d

- If the process is non-stationary then first differences of the series are computed to determine if that operation results in a stationary series.
- The process is continued until a stationary time series is found.
- This then determines the value of d.

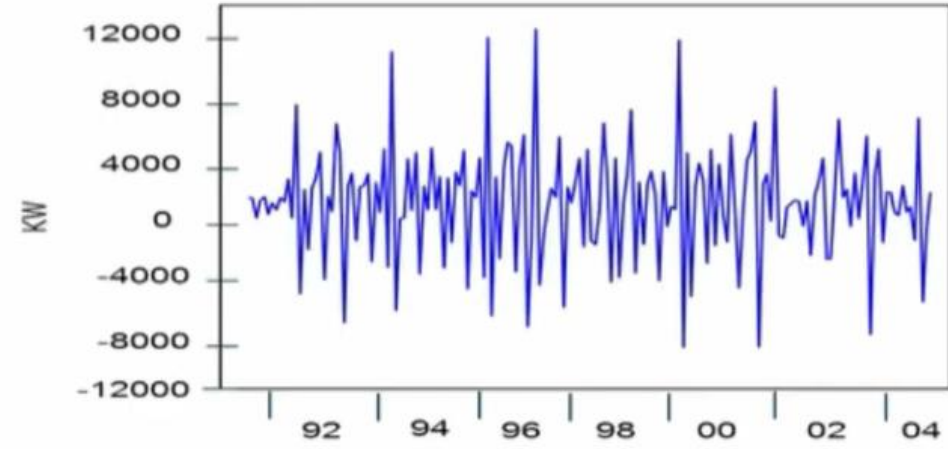
Consumption of electricity in a particular area across years

Plot of actual values v/s Time



Non - Stationary Series

Plot of differenced values v/s Time



Stationary Series

Components of ARIMA

Three components of ARIMA – AR, I and MA

MA

Moving average part accounts for the random component of the series whether it has some pattern in random part

q

$q = 0/1/2/.../q :$
Observations depend on 'q' previous residuals

Components of ARIMA

Three components of ARIMA – AR, I and MA

MOVING AVERAGE (MA) MODELS

Period	Actual	Forecast	Error
1	Y1	F1	$\varepsilon_1 = A_1 - F_1$
2	Y2	F2	$\varepsilon_2 = A_2 - F_2$
3	Y3	F3	$\varepsilon_3 = A_3 - F_3$

$$Y_t = \mu + \varepsilon_t - \omega_1 \varepsilon_{t-1} - \omega_2 \varepsilon_{t-2} - \dots - \omega_q \varepsilon_{t-q}$$

MA(q) model

ARMA MODELS

Period	Actual	Forecast	Error
1	Y1	F1	$\varepsilon_1 = A_1 - F_1$
2	Y2	F2	$\varepsilon_2 = A_2 - F_2$
3	Y3	F3	$\varepsilon_3 = A_3 - F_3$

$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \varepsilon_t \\ - \omega_1 \varepsilon_{t-1} - \omega_2 \varepsilon_{t-2} - \dots - \omega_q \varepsilon_{t-q}$$

ARMA(p, q) model

ARIMA

STATIONARY DATA

AR(p)

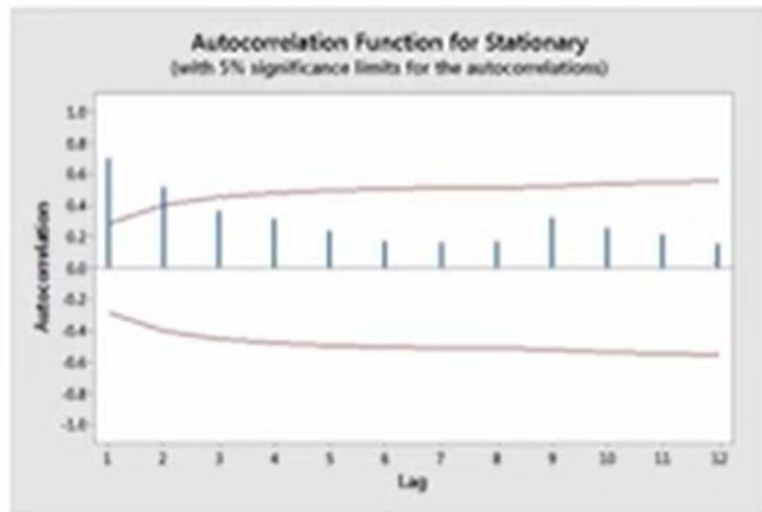
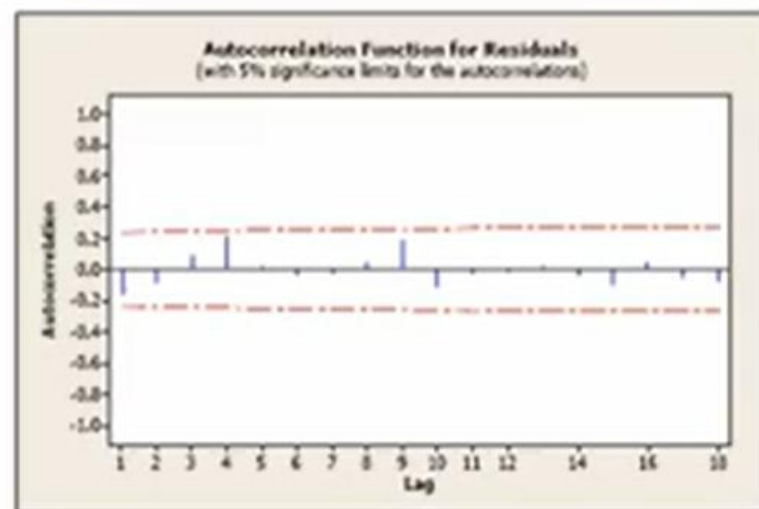
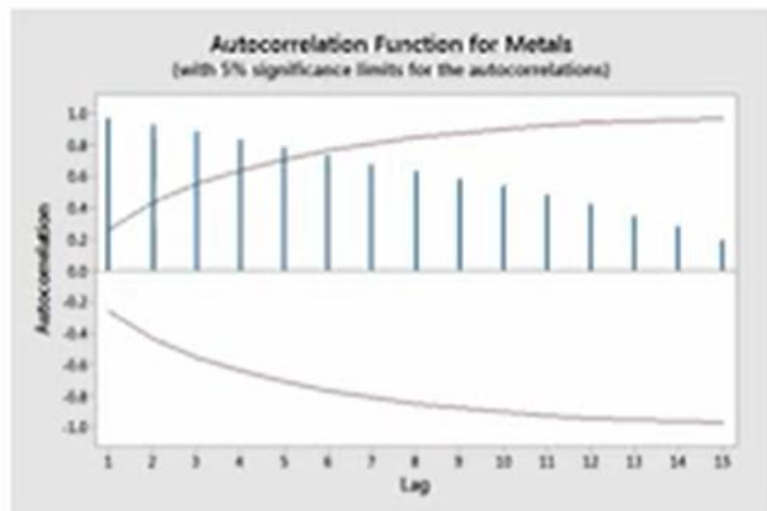
MA(q)

ARMA(p, q)

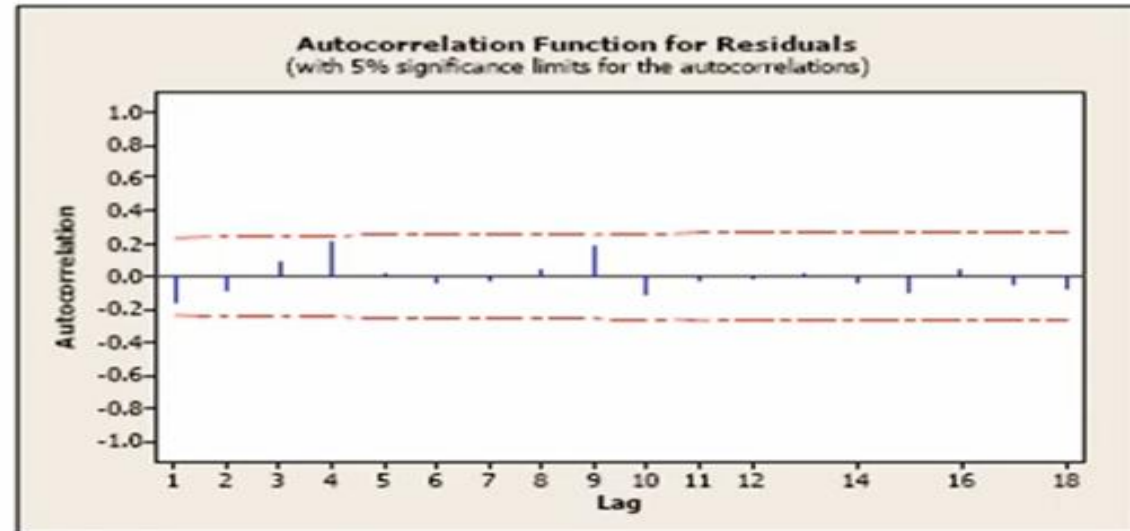
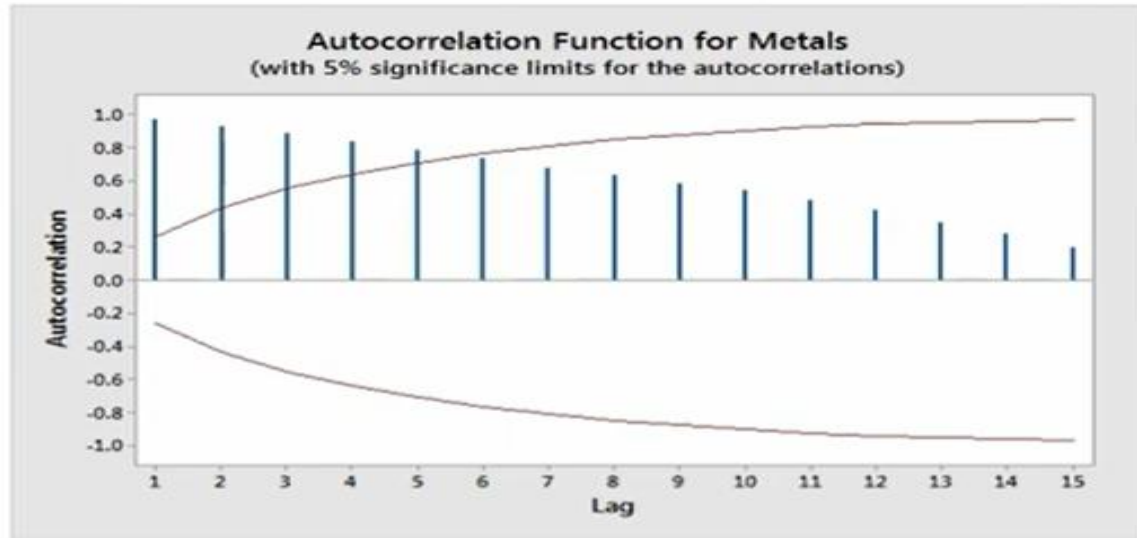
AUTO-REGRESSIVE INTEGRATED MOVING AVERAGE MODELS

ARIMA(p, d, q)

STEP-1: CHECK STATIONARITY

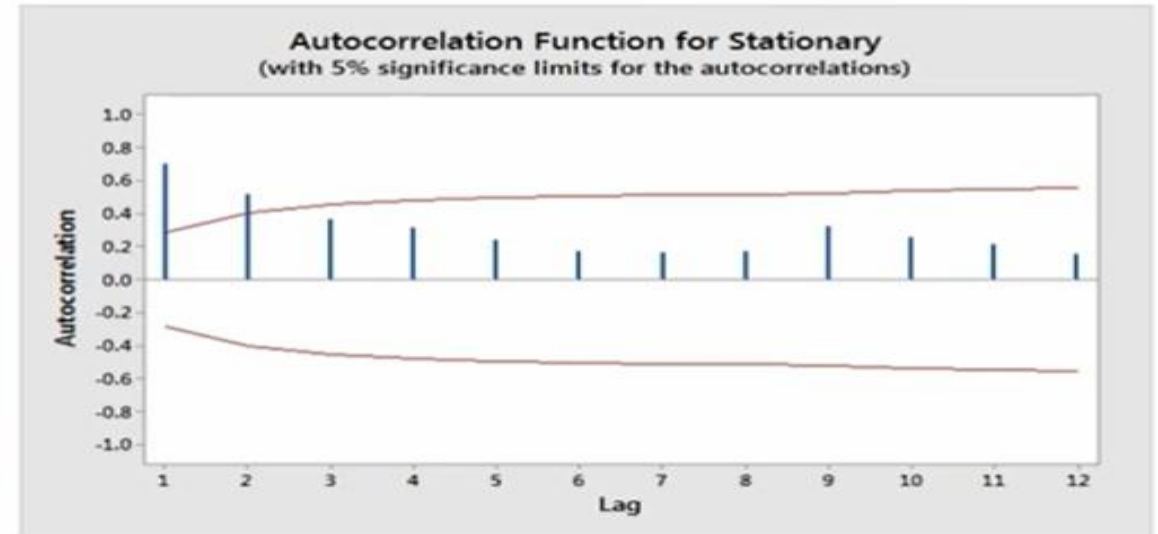


Non-stationary (die out)

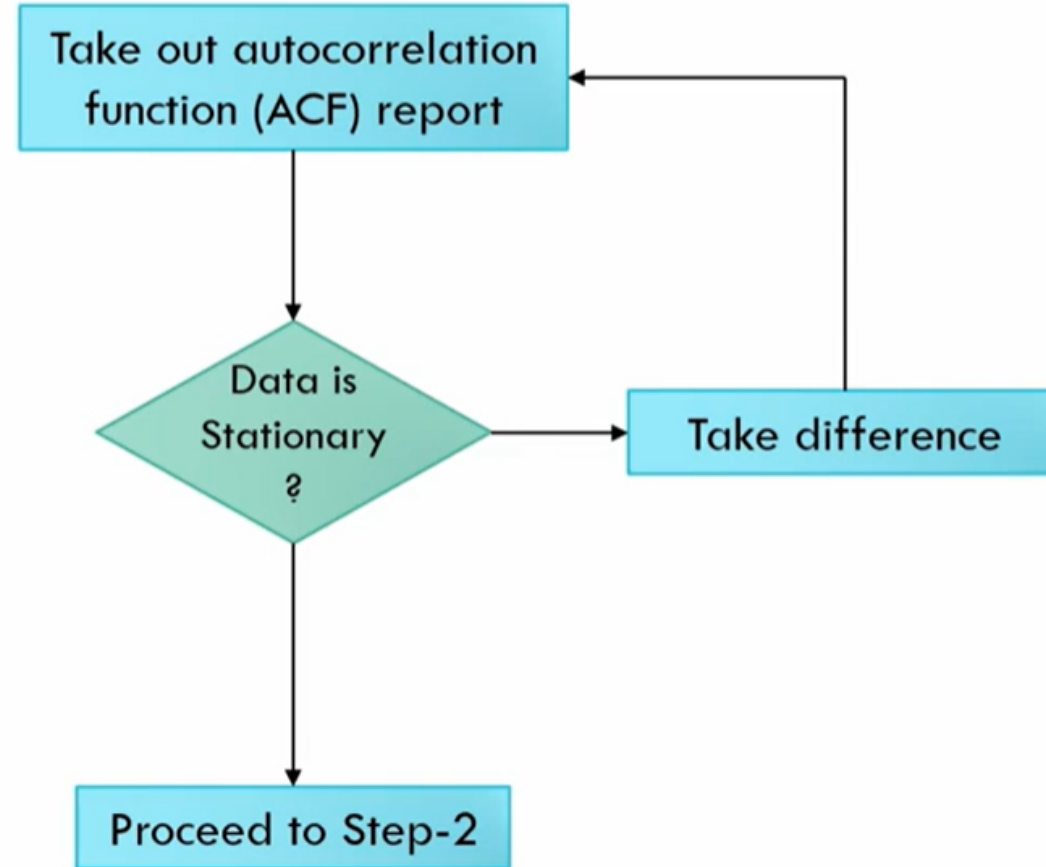


Random (white noise)

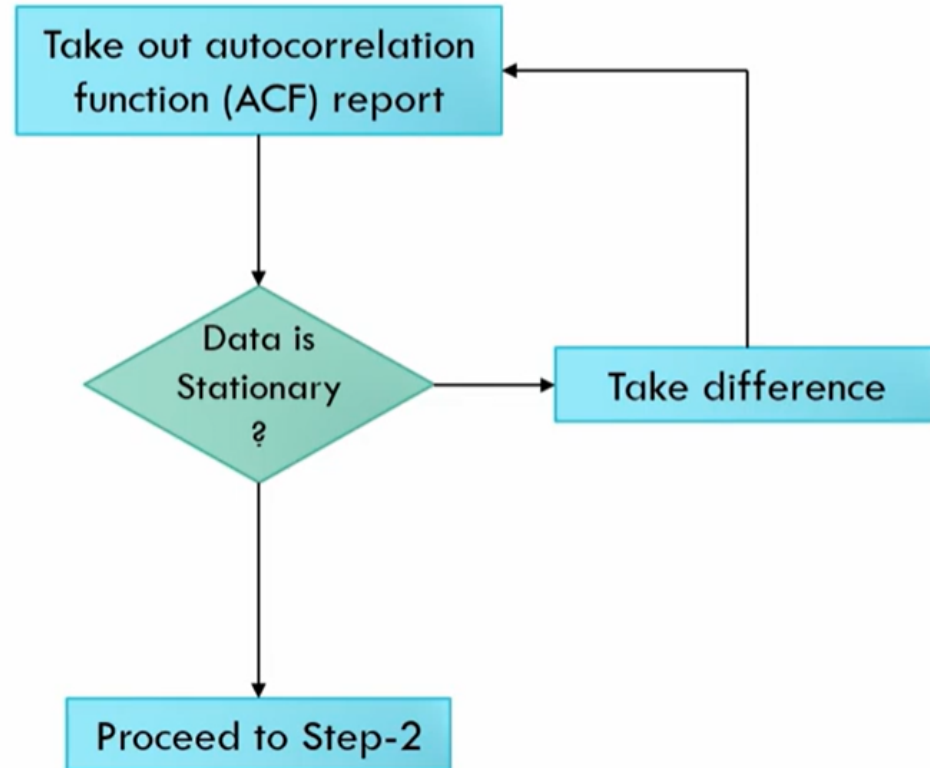
Stationary (cut off)



STEP-1: CHECK STATIONARITY



STEP-1: CHECK STATIONARITY



Data	First Difference	Second difference
y1		
y2	$y2 - y1$	
y3	$y3 - y2$	$(y3 - y2) - (y2 - y1)$
y4	$y4 - y3$	$(y4 - y3) - (y3 - y2)$
y5	$y5 - y4$	$(y5 - y4) - (y4 - y3)$
y6	$y6 - y5$	$(y6 - y5) - (y5 - y4)$
y7	$y7 - y6$	$(y7 - y6) - (y6 - y5)$

STEP-2: FIND POSSIBLE MODELS

Take out autocorrelation function (ACF) and partial autocorrelation function (PACF) report

$AR(p)$

ACF – Die out

PACF – Cut off after p-th lag

$MA(q)$

ACF – Cut off after q-th lag

PACF – Die out

$ARMA(p,q)$

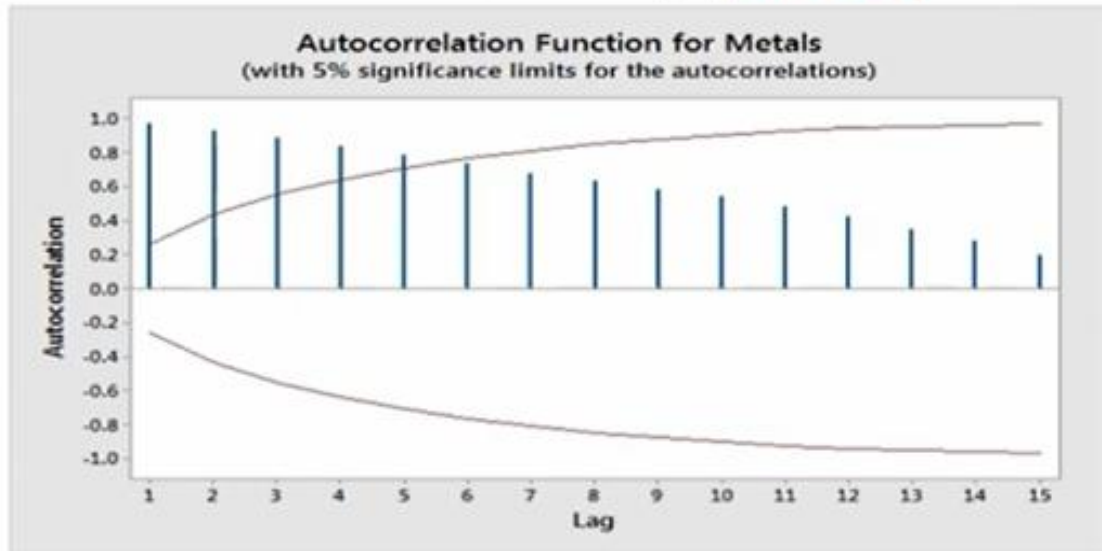
ACF – Die out

PACF – Die out

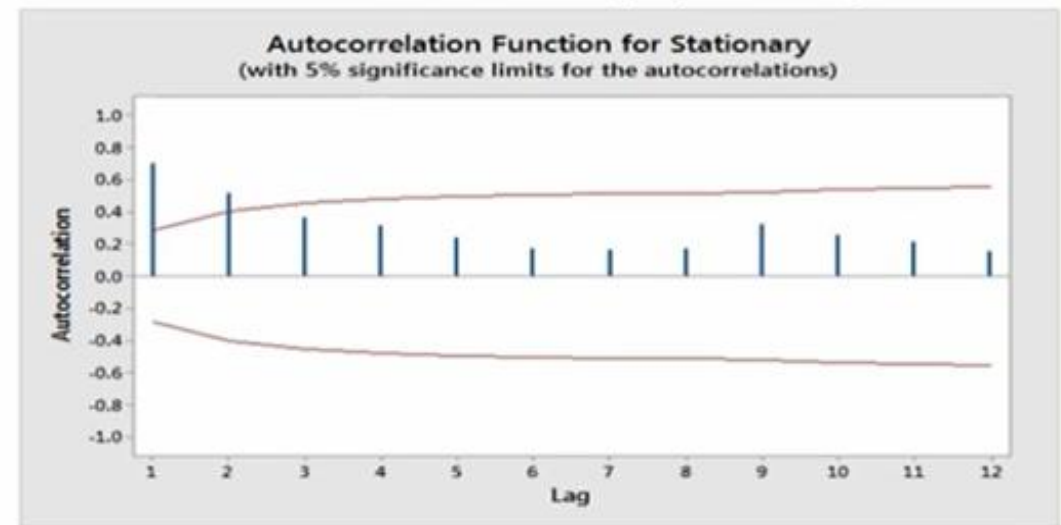
STEP-2: FEW ILLUSTRATIONS

	ACF	PACF
MA(q)	Cut off after q-th lag	Die out
AR(p)	Die out	Cut off after p-th lag
ARMA(p,q)	Die out	Die out

Non-stationary (die out)



Stationary (cut off)



STEP-2: FEW ILLUSTRATIONS

	ACF	PACF
MA(q)	Cut off after q-th lag	Die out
AR(p)	Die out	Cut off after p-th lag
ARMA(p,q)	Die out	Die out

ACF	PACF	Model choice
Cut off after 1 st lag	Die out	MA(1)
Die out	Cut off after 1 st lag	AR(1)
Die out	Die out	ARMA(1,1)
Cut off after 1 st lag	Cut off after 1 st lag	MA(1), AR(1), ARMA(0,1), ARMA(1,0)

STEP-2: FEW ILLUSTRATIONS

	ACF	PACF
MA(q)	Cut off after q-th lag	Die out
AR(p)	Die out	Cut off after p-th lag
ARMA(p,q)	Die out	Die out

ACF	PACF	Model choice
Cut off after 1 st lag	Die out	MA(1)
Die out	Cut off after 1 st lag	AR(1)
Die out	Die out	ARMA(1,1)
Cut off after 1 st lag	Cut off after 1 st lag	MA(1), AR(1), ARMA(0,1), ARMA(1,0)

MA(1) → ARIMA(0,1,1)

AR(1) → ARIMA(1,1,0)

ARMA(1,1) → ARIMA(1,1,1)

If differencing is done as part of step 1 to bring in stationary and d=1 gave u stationary then use ARIMA

STEP-3: ENTER THE DATA AND TAKE OUT REPORT

ARIMA

Series:

☐ Fit seasonal model
Period: 12

	Nonseasonal	Seasonal
Autoregressive:	0	0
Difference:	0	0
Moving average:	0	0

☒ Include constant term in model

☐ Starting values for coefficients:

Select

Graphs... Forecasts... Results... Storage... Help OK Cancel

Entry form in Minitab

Report

ARIMA Model: Trade

Final Estimates of Parameters

Type		Coef	SE Coef	T-Value	P-Value
AR	1	-0.504	0.114	-4.42	0.000
Constant		150.415	0.325	463.34	0.000
Mean		100.000	0.216		

Number of observations: 60

Residual Sums of Squares

DF	SS	MS
58	366.733	6.32299

Back forecasts excluded

Modified Box-Pierce (Ljung-Box) Chi-Square Statistic

Lag	12	24	36	48
Chi-Square	4.05	12.13	25.62	32.09
DF	10	22	34	46
P-Value	0.945	0.955	0.849	0.940

STEP-4: INTERPRETATION OF REPORT AND FUTURE FORECAST

ARIMA Model: Trade

Final Estimates of Parameters

Type		Coef	SE Coef	T-Value	P-Value
AR	1	-0.504	0.114	-4.42	0.000
Constant		150.415	0.325	463.34	0.000
Mean		100.000	0.216		

Number of observations: 60

Residual Sums of Squares

DF	SS	MS
58	366.733	6.32299

Back forecasts excluded

Modified Box-Pierce (Ljung-Box) Chi-Square Stati

Lag	12	24	36	48
Chi-Square	4.05	12.13	25.62	32.09
DF	10	22	34	46
P-Value	0.945	0.955	0.849	0.940

Interpretation:

1. Residual plot (normality check and time sequence plot)
2. Residual autocorrelation should be within $\pm 2/\sqrt{n}$.
3. The P-value corresponding to the slope < 0.05 – for significance
4. Ljung-Box p-values should be > 0.05 – overall model adequacy check

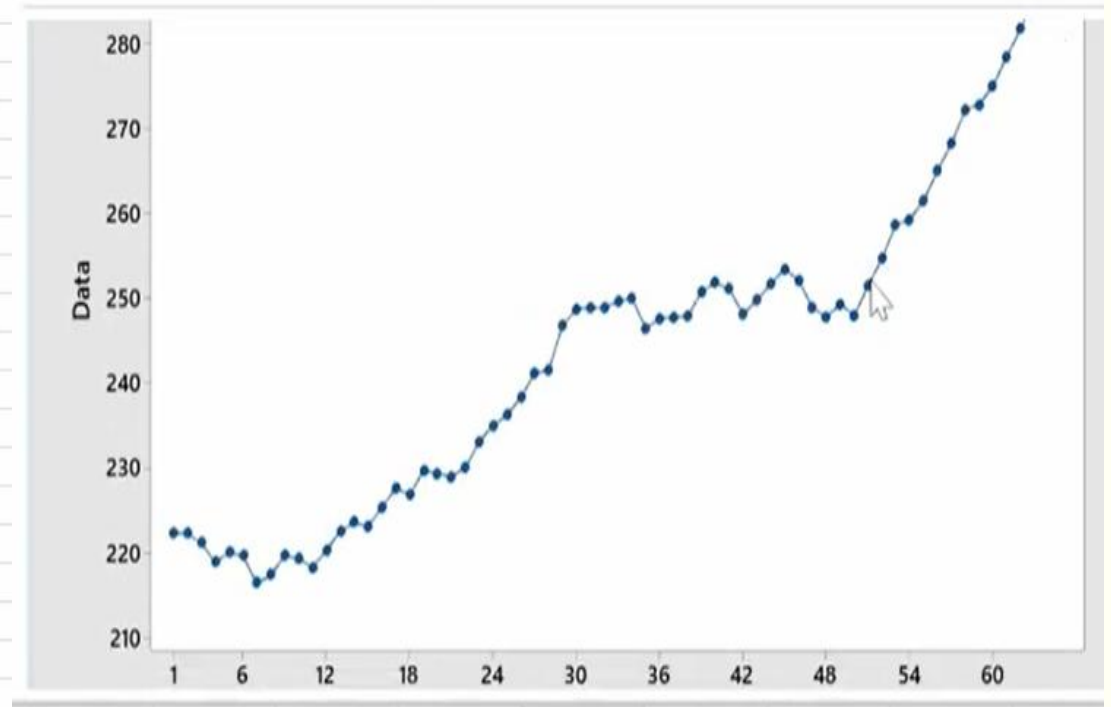
ARIMA

	C1	C2	C3	C4	C5	C6	C7	C8	C9
	Period	Data							
1	1	222.3415							
2	2	222.2416							
3	3	221.1717							
4	4	218.8818							
5	5	220.0519							
6	6	219.6120							
7	7	216.4021							
8	8	217.3322							
9	9	219.6923							
10	10	219.3224							
11	11	218.2525							
12	12	220.3026							
13	13	222.5427							
14	14	223.5628							
15	15	223.0729							
16	16	225.3630							
17	17	227.6031							
18	18	226.8232							
19	19	229.6933							
20	20	229.3034							
21	21	228.9535							
22	22	229.9936							
23	23	233.0537							

Step-1: Check Stationarity

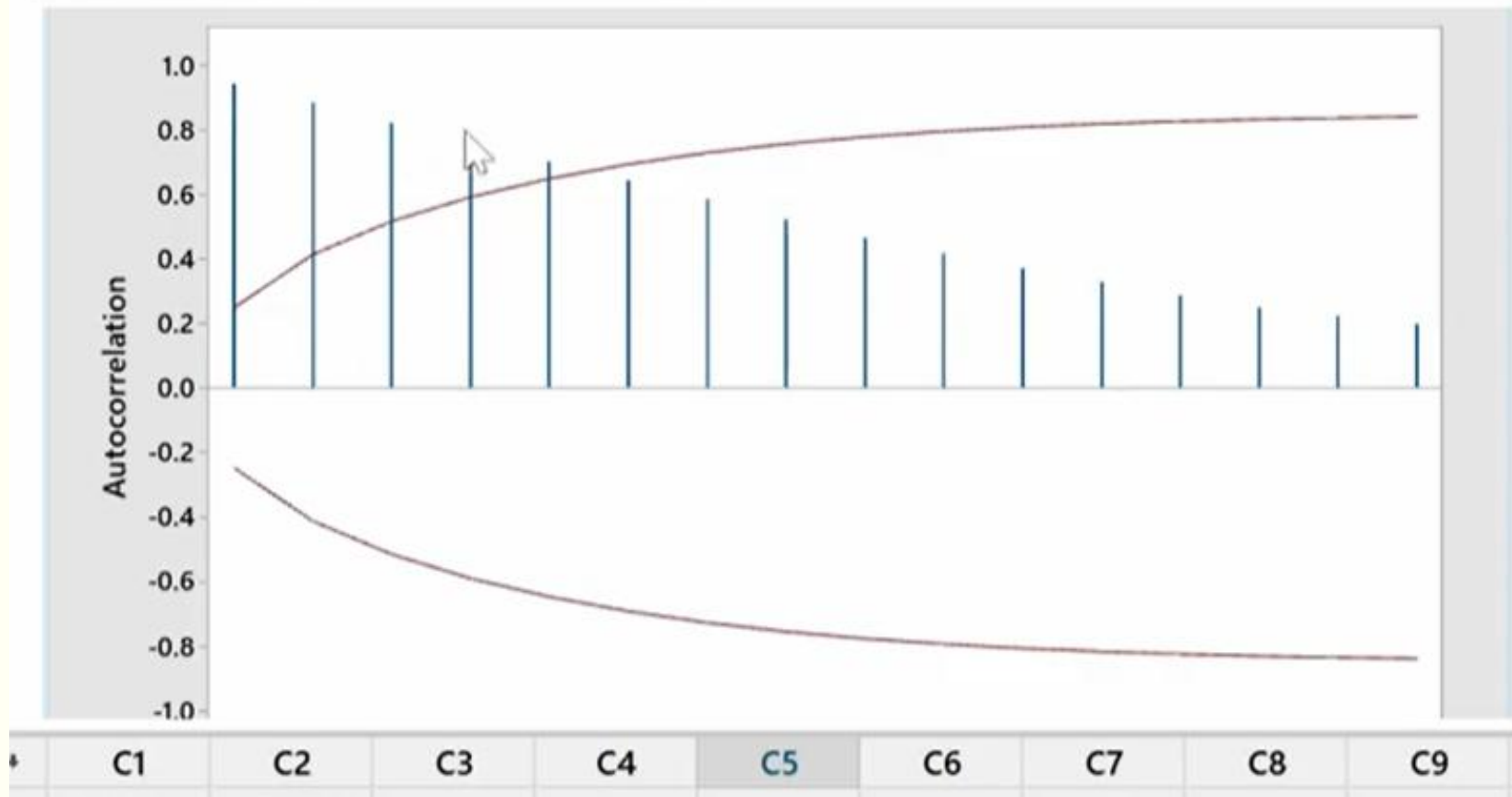
WORKSHEET 1

Time Series Plot of Data



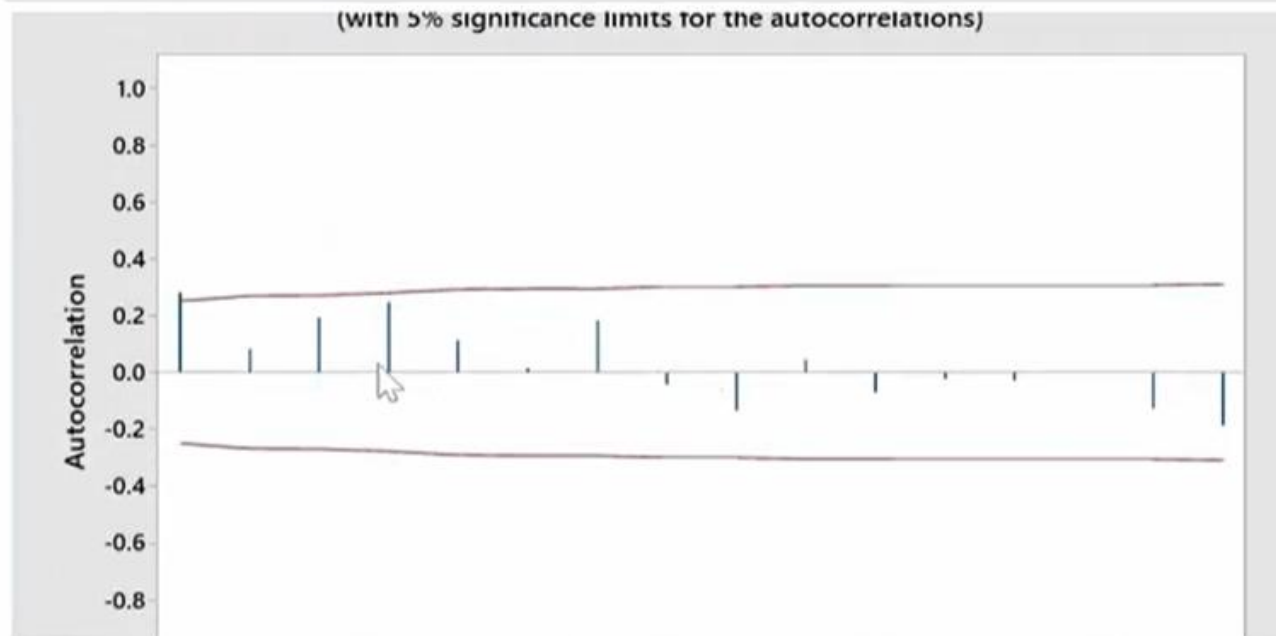
ARIMA

Autocorrelation Function: Data



ARIMA after differencing d=1

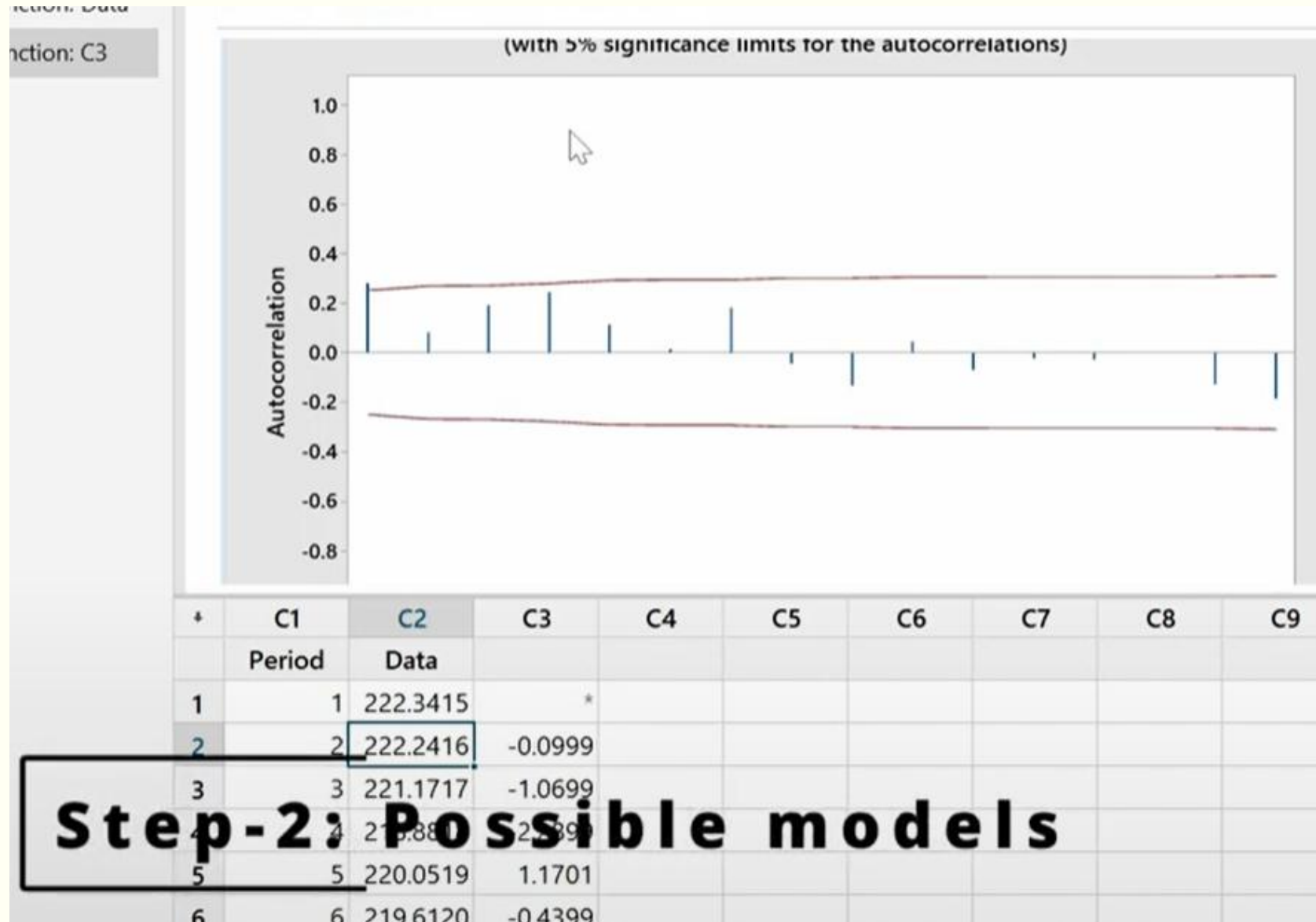
Autocorrelation Function: C3



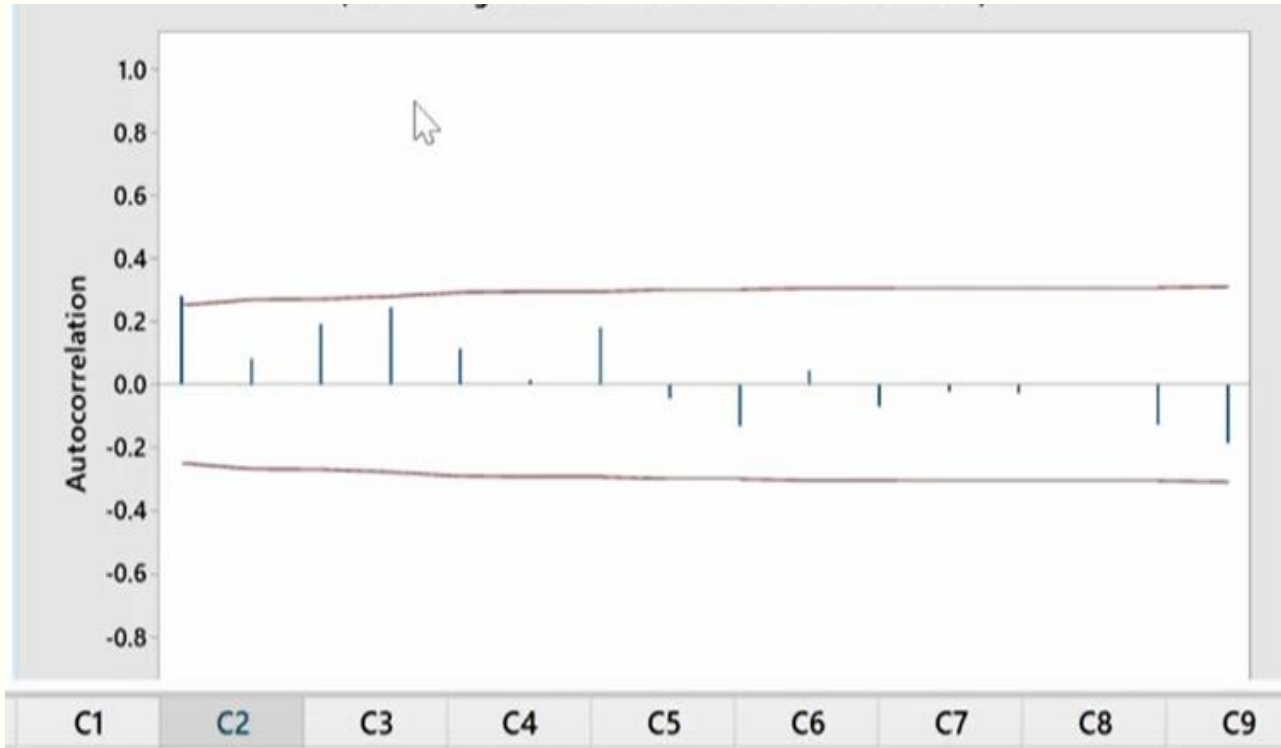
	C1	C2	C3	C4	C5	C6	C7	C8	C9
	Period	Data							
1	1	222.3415	*						
2	2	222.2416	-0.0999						
3	3	221.1717	-1.0699						
4	4	218.8818	-2.2899						
5	5	220.0519	1.1701						
6	6	219.6120	-0.4399						

Worksheet 1

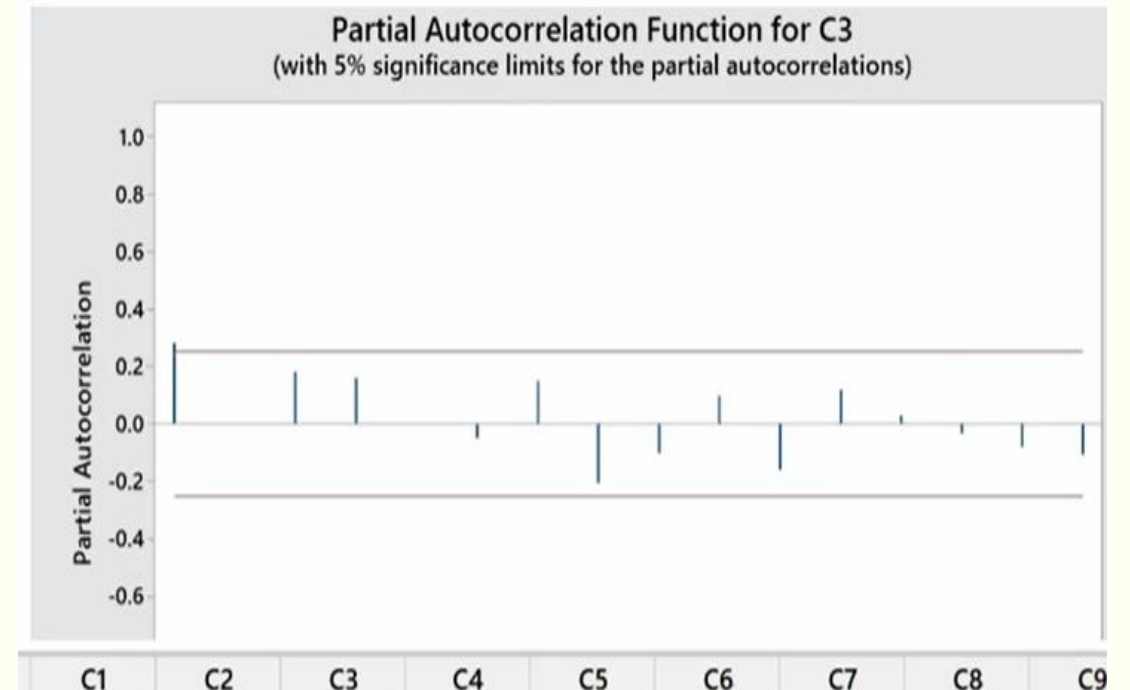
ARIMA



ARIMA after differencing d=1



Partial Autocorrelation Function: C3



STEP-2: FEW ILLUSTRATIONS

	ACF	PACF
MA(q)	Cut off after q-th lag	Die out
AR(p)	Die out	Cut off after p-th lag
ARMA(p,q)	Die out	Die out

ACF	PACF	Model choice
Cut off after 1 st lag	Die out	MA(1)
Die out	Cut off after 1 st lag	AR(1)
Die out	Die out	ARMA(1,1)
Cut off after 1 st lag	Cut off after 1 st lag	MA(1), AR(1), ARMA(0,1), ARMA(1,0)

MA(1) → ARIMA(0,1,1)

AR(1) → ARIMA(1,1,0)

ARMA(1,1) → ARIMA(1,1,1)

If differencing is done as part of step 1 to bring in stationary and d=1 gave u stationary then use ARIMA

ARIMA

ARIMA(1,1,0)

ARIMA(0,1,1)

ARIMA

Time Series Plot of Data

Autocorrelation Function: Data

Autocorrelation Function: C3

Partial Autocorrelation Function: C3

ARIMA Model: Data

ARIMA Model: Data

WORKSHEET 1

ARIMA Model: Data

Final Estimates of Parameters

Type	Coef	SE Coef	T-Value	P-Value
MA 1	-0.291	0.123	-2.37	0.021
Constant	1.038	0.303	3.42	0.001

Differencing: 1 regular difference

Number of observations: Original series 65, after differencing 64

Residual Sums of Squares

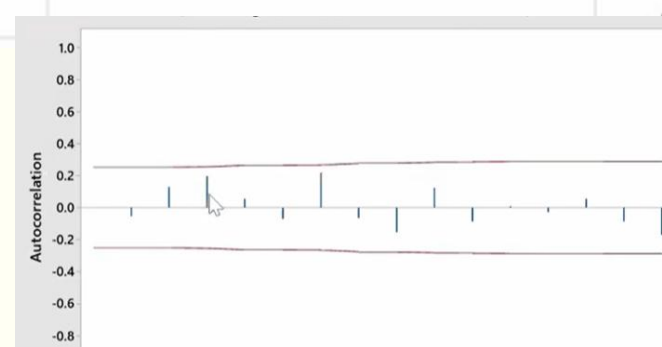
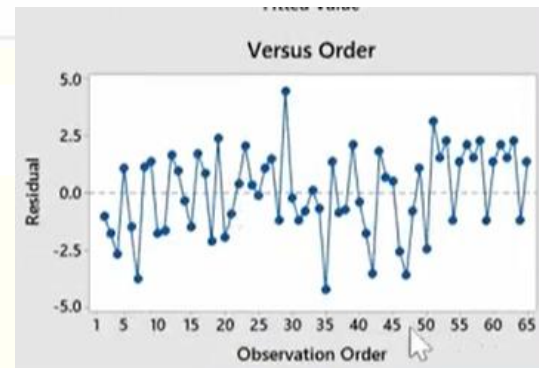
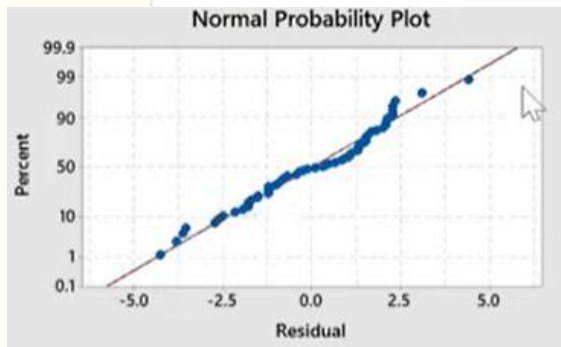
DF	SS	MS
62	219.339	3.53773

* Period	C1	C2	C3	C4	C5	C6	C7	C8	C9
1	1	222.3415	*						
2	2	222.2416	-0.0999						
3	3	221.1717	-1.0699						
4	4	217.8815	-2.819						
5	5	220.0519	1.1701						

Step-4: Select the best model

ARIMA

	ARIMA(1,1,0)	ARIMA(0,1,1)
Residual Normality		
Residual time sequence		
Residual autocorrelation (within $\pm 2/\sqrt{n}$)		
P-value of the regression line		
P-values of the Ljung-Box statistics		
MS value		



Final Estimates of Parameters

Type	Coef	SE Coef	T-Value	P-Value
AR 1	0.284	0.122	2.33	0.023
Constant	0.741	0.235	3.15	0.001

Modified Box-Pierce (Ljung-Box) Chi-Square Statistic

Lag	12	24	36	48
Chi-Square	11.83	29.15	37.12	48.09
DF	10	22	34	46
P-Value	0.297	0.141	0.327	0.388

Residual Sums of Squares

DF	SS	MS
62	219.209	3.53564

Back forecasts excluded

ARIMA

	AR(1) ARIMA(1,1,0)	MA(1) ARIMA(0,1,1)
Residual Normality	Normal	Normal
Residual time sequence	No trend and random	No trend and random
Residual autocorrelation (within $\pm 2/\sqrt{n}$)	$< 2/\sqrt{n}$	$< 2/\sqrt{n}$
P-value of the regression line	$0.023 < 0.05$	$0.021 < 0.05$
P-values of the Ljung-Box statistics	> 0.05	> 0.05
MS value	3.53564	3.53773

ARIMA

Forecasts from period 65

Period	Forecast	95% Limits		Actual
		Lower	Upper	
66	289.948	286.262	293.634	
67	291.080	285.080	297.081	
68	292.143	284.312	299.974	
69	293.187	283.832	302.541	
70	294.224	283.550	304.898	

ARIMA

A class of statistical model for analyzing and forecasting time series data

ARIMA - **A**uto**R**egressive **I**ntegrated **M**oving **A**verage

Data shows evidence of non-stationarity

A random variable that is a time series is stationary if its statistical properties are all constant over time.

ARIMA

A stationary series has **no trend**, its variations around its mean have a **constant amplitude**, and it wiggles in a consistent fashion

The latter condition means that its **autocorrelations** remain constant over time

A random variable of this form can be viewed as a combination of **signal and noise**

An ARIMA model can be viewed as a “filter” that tries to separate the signal from the noise, and the signal is then extrapolated into the future to obtain forecasts.

ARIMA

AR = AutoRegressive - uses the dependent relationship between an observation and some number of lagged observations.

I = Integrated - The use of differencing of raw observations

MA = Moving Average - uses the dependency between an observation and residual errors from a moving average model applied to lagged observations.

ARIMA

AR = AutoRegressive - uses the dependent relationship between an observation and some number of lagged observations.

p = Lag order.

I = Integrated - The use of differencing of raw observations

d = Degree of differencing

MA = Moving Average - uses the dependency between an observation and residual errors from a moving average model applied to lagged observations.

q = Order of moving average

ARIMA

Introduction



ARMA and ARIMA models

- ARMA: Autoregressive Moving Average
- AR(I)MA: Autoregressive Integrated Moving Average

Box and Jenkins (1970) introduced a three steps method to select appropriate models for estimating and forecasting univariate models.

- "Let my past predict you my future".

Box and Jenkins three stages are:

- Identification
- Estimation
- Diagnostic and Forecasting

Module No: -5 Time Series Forecasting

5.1 Taxonomy of Time Series Forecasting methods,

5.2 Time Series Decomposition

5.3 Smoothing Methods:

Average method,

Moving Average smoothing,

5.4 Time series analysis using linear regression,

5.5 ARIMA Model,

5.6 Performance Evaluation:

Mean Absolute Error,

Root Mean Square Error,

Mean Absolute Percentage Error,

Mean Absolute Scaled Error

5.6 Performance Evaluation

$$e_i = y_i - \hat{y}_i$$

The forecast error is the difference between the actual value y_i and the forecasted value $\hat{y}_i$.

$$\text{Mean absolute error} = \text{mean}(|e_i|)$$

$$\text{Root mean squared error} = \sqrt{\text{mean}(e^2)}$$

$$\text{Mean absolute percentage error} = \text{mean}(|p_i|)$$

$$p_i = 100 \frac{e_i}{y_i}$$

FORECAST ERROR ESTIMATES

MFE

Mean Forecast Error

$$\text{MFE} = \frac{\sum_{t=1}^n (A_t - F_t)}{n}$$

MAD

Mean Absolute Deviation

$$\text{MAD} = \frac{\sum_{t=1}^n |A_t - F_t|}{n}$$

MSE

Mean Squared Error

$$\text{MSE} = \frac{\sum_{t=1}^n (A_t - F_t)^2}{n}$$

MAPE

Mean Absolute Percentage Error

$$\text{MAPE} = \frac{\sum_{t=1}^n \left| \frac{A_t - F_t}{A_t} \right|}{n} \times 100$$

MPE

Mean Percentage Error

$$\text{MPE} = \frac{\sum_{t=1}^n \frac{A_t - F_t}{A_t}}{n} \times 100$$

TS

Tracking Signal

$$\text{TS} = \frac{\sum_{t=1}^n (A_t - F_t)}{\text{MAD}}$$

ACCURACY



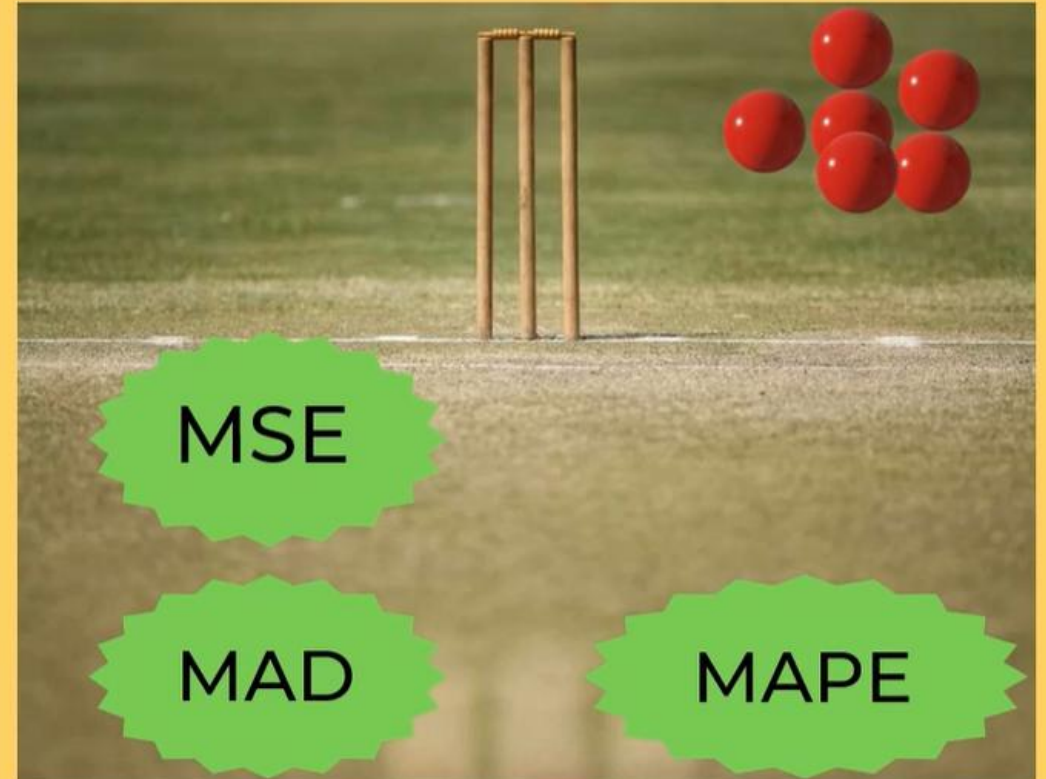
MFE

MPE

ABSOLUTE

RELATIVE

PRECISION



MSE

MAD

MAPE

ABSOLUTE

RELATIVE

Be
Good
Do
Good

**To be good, and to
do good, is all
we have to do.**
